



DEPARTMENT OF PUBLIC HEALTH AND ENVIRONMENT

Air Quality Control Commission

REGULATION NUMBER 27

GREENHOUSE GAS EMISSIONS AND ENERGY MANAGEMENT FOR MANUFACTURING

5 CCR 1001-31

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

Outline of Regulation

PART A	General Provisions
PART B	GEMM 2 Facility Requirements
PART C	Energy-Intensive Trade-Exposed Stationary Source Requirements
PART D	Greenhouse Gas Credit Trading
PART E	Statements of Basis, Specific Statutory Authority and Purpose

Pursuant to Colorado Revised Statutes § 24-4-103 (12.5), materials incorporated by reference are available for public inspection during normal business hours, or copies may be obtained at a reasonable cost from the Air Quality Control Commission (the Commission), 4300 Cherry Creek Drive South, Denver, Colorado 80246-1530. The material incorporated by reference is also available through the United States Government Printing Office, online at www.govinfo.gov. Materials incorporated by reference are those editions in existence as of the date indicated and do not include any later amendments.

PART A General Provisions

I. Purpose and Applicability

- I.A. The purpose of this regulation is to require manufacturing stationary sources to reduce greenhouse gas emissions pursuant to § 25-7-105(1)(e), C.R.S., (2023).
- I.B. This regulation applies to manufacturing stationary sources with annual direct emissions equal to or greater than 25,000 metric tons of CO₂e per year, as reported pursuant to Regulation Number 22, Part A. Once a manufacturing stationary source reports annual direct GHG emissions equal to or greater than 25,000 metric tons of CO₂e per year, as reported pursuant to Regulation Number 22, Part A, or U.S. Environmental Protection Agency's (EPA) Greenhouse Gas Reporting Program (Title 40, Part 98, of the Code of Federal Regulations (CFR)) (Part 98), on or after the year 2015, the requirements of Regulation Number 27 continue to apply even if the source reports less than 25,000 metric tons of CO₂e in direct GHG emissions for any year thereafter.
- I.C. If any section, clause, phrase, or standard contained in this regulation is for any reason held to be inoperative, unconstitutional, void, or invalid, the validity of the remaining portions thereof will not be affected and the Commission declares that it severally passed and adopted these provisions separately and apart.

II. Definitions

- II.A. "2015 GHG emissions" means the direct emissions reported in metric tons of CO₂e by a GEMM 2 facility to the EPA's Greenhouse Gas Reporting Program (40 CFR, Part 98, Subparts C through JJ), for calendar year 2015.
- II.B. "2030 social cost of GHGs" means for carbon dioxide, \$89 per metric ton of carbon dioxide; for methane, \$2,500 per metric ton of methane; for nitrous oxide, \$33,000 per metric ton of nitrous oxide; and for all other greenhouse gases, the corresponding cost of CO₂e.
- II.C. "Additional emissions reductions" means GHG emission reductions that exceed any GHG emission reductions otherwise required by law, regulation, or legally binding mandate.
- II.D. "Alternate account representative" means an individual designated pursuant to Part D, Section II.C. to take actions on the manufacturing stationary source' accounts.
- II.E. "Annual emissions limitation" means the number of metric tons of CO₂e an EITE stationary source may emit as calculated in Part C, Section III.A.1.
- II.F. "Auction" means the process of creating a market for the sale of GHG credits by taking bids from potential GHG credit buyers, taking offers from potential GHG credit sellers, determining which GHG credits buyers purchased, determining how many GHG credits will be sold and to which buyers, and determining the auction settlement price. The auction will not collect payment from winning bidders but will instead instruct buyers to which sellers they must direct payment.
- II.G. "Auction administrator" means the Division or the Division's agent charged with administering an annual auction for the voluntary sale of GHG credits.
- II.H. "Auction settlement price" means the price announced by the auction administrator at the conclusion of each annual auction pursuant to Part D, Section IV.
- II.I. "Audit plan" means the proposed audit scope, timelines and team submitted by the EITE stationary source to the Division for approval pursuant to Part C.
- II.J. "Audit report" means the resulting document from the audit containing all the information and data required under Part C.
- II.K. "Audit scope" means the GHG emission units and the energy consumption sources included in the energy and emissions control audit and identified in the approved audit plan.
- II.L. "Audit team" means one or more persons performing the audit. The audit team must consist of at least one qualified third-party auditor. Additional capabilities and knowledge of the audit team must include, but are not limited to, technical expertise with specific operating and maintenance practices for the industry being audited; expertise in conducting GHG and energy management system audits; and expertise of the EITE stationary source's domestic and international market. The audit team must include individual(s) with documented audit expertise in the relevant industrial sector.
- II.L.1. A bachelor's level college degree or equivalent in science, technology, business, statistics, math, environmental policy, economic, or financial auditing; or evidence demonstrating completion of significant and relevant work experience or other personal development activities that have provided the applicant with the communication, technical, and analytical skills to conduct audit; and

- II.L.2. Sufficient workplace experience to act as an auditor, including a minimum two years of full-time experience in a professional role that involved emissions data management, emissions technology, emissions inventories, environmental auditing or other technical skills necessary to conduct the audit.
- II.M. “Carbon dioxide equivalent” (CO₂e) means a standard used to compare the emissions from various greenhouse gases based upon their global warming potential (GWP). The CO₂e is determined by multiplying the mass amount of emissions (metric tons per year), for each GHG constituent by that gas’s GWP, codified in 40 CFR Part 98, Subpart A, Table A-1 (as of December 11, 2014), and summing the resultant values to determine CO₂e (metric tons per year).
- II.N. “Certification body” means a professional organization that has been accredited to issue lead auditor certifications for a specific sector or to a specific standard.
- II.O. “Co-benefits” means the additional benefits associated with the reduction of harmful air pollutants to local communities, including localized air quality benefits.
- II.P. “Colorado EnviroScreen” means Colorado’s interactive environmental justice mapping tool, which compiles 35 environmental, health, and demographic indicators to identify and visualize areas with higher environmental health risks, as of the effective date of the rule. The tool also shows places that meet the statutory definition of a disproportionately impacted community in § 24-4-109(2)(b)(II), C.R.S. (May 23, 2023).
- II.Q. “Combined heat and power unit” (also known as a “cogeneration unit”) means a unit that simultaneously produces both electric power and useful thermal output from the same primary energy source, and may include facilities where electricity is generated from waste steam or is generated by a stationary combustion turbine.
- II.R. “Compliance year” means any year in which a GEMM 2 facility must comply with a GEMM 2 annual GHG emissions requirement.
- II.S. “Credit account” means an account for a manufacturing stationary source that is created by the Division or its agent, to which the Division and/or the manufacturing stationary source transfers GHG credits to meet the manufacturing stationary source’s compliance obligations.
- II.T. “Direct GHG emissions” means GHG emissions from a manufacturing stationary source that are reported to the State of Colorado under Regulation Number 22, Part A and/or to the EPA under Part 98 and measured in terms of CO₂e.
- II.U. “Disproportionately impacted community” means those communities that meet the definition contained in § 24-4-109(2)(b)(II), C.R.S. (May 23, 2023). For purposes of Regulation Number 27, disproportionately impacted community means any census block group identified in the disproportionately impacted community layer in the most recent version of Colorado EnviroScreen as of the date of the effective rule.
- II.V. “Energy and GHG emission control audit” (the audit) means a rigorous examination of the GHG emissions and energy consumption of an EITE stationary source with the goal of analyzing and recommending GHG BAECT and energy BMPs, and identifying opportunities for reduction in GHG emissions and energy consumption for the facility, conducted consistent with the requirements set forth in this section.

- II.W. “Energy best management practices” (energy BMPs) means the best energy efficiency practices available to the EITE stationary source, based on the maximum degree of energy efficiency that is achievable on a case-by-case basis, taking into account energy, environmental, and economic impacts, and is achievable for such facility through application of production process improvements and available equipment or process control methods, systems, and techniques, and includes incorporating all the key elements of strategic energy management (SEM), such that the facility continually improves its energy performance, reduces energy costs, and reduces GHG emissions associated with energy use.
- II.X. “Energy efficiency” means using less electricity or fuel to produce the same quantity of product or service.
- II.Y. “Energy-intensive, trade-exposed manufacturing stationary source” (EITE stationary source) means a source that principally engages in cement and concrete product manufacturing, NAICS code 3273; foundries, NAICS code 3315; iron and steel mills and ferroalloy manufacturing, NAICS code 3311; and/or pulp, paper, and paperboard mills, NAICS code 3221.
- II.Z. “Federal ENERGY STAR ® Program” means the EPA’s voluntary program for industrial manufacturers through which specific energy performance indicators are measured and compared across industries and to which facilities are certified if they are achieving an Energy Star scoring of 75 or greater.
- II.AA. “Gas distribution utility” means a public utility providing gas service to more than ninety thousand retail customers. Gas distribution utility does not include a municipal gas distribution utility.
- II.BB. “GEMM 2 facility” means a stationary source located in Colorado that principally engages in manufacturing activities and directly emits equal to or greater than 25,000 metric tons per year of CO₂e emissions, as reported pursuant to Regulation Number 22; including any existing EITE stationary source that emits equal to or greater than 25,000 metric tons per year of CO₂e emissions and elects to be regulated under Part B. Manufacturing activities include the mechanical, physical, or chemical transformation of materials, substances, or components into new products. This is limited to facilities with NAICS codes beginning with 31-33.
- II.CC. “GEMM 2 facility GHG baseline emissions” means, for stationary sources qualifying as GEMM 2 facilities as of January 1, 2023, the higher reported direct GHG emissions from either the 2021 or 2022 calendar year, measured in metric tons of CO₂e as reported pursuant to Regulation Number 22, Part A., as revised to correct any previous inaccuracies or to account for capital investments between 2015 and 2021 that increased a GEMM 2 facility’s production capacity by over thirty (30) percent, for which additional production, as of 2022, was not yet realized. If a GEMM 2 facility met the criteria for the production-based adjustment, the GEMM 2 facility’s baseline was revised upwards to account for seventy-five (75) percent of the GHG emissions increase resulting from the production capacity expansion, provided, however, that if the GEMM 2 facility met the requirements of Part B, Section I.A.1, its GEMM 2 facility baseline was revised upwards to account for one-hundred (100) percent of the GHG emissions increase resulting from the production capacity expansion.
- II.DD. “GEMM 2 annual GHG emissions requirement” means for each stationary source qualifying as a GEMM 2 facility as of January 1, 2023, the calculated CO₂e emissions requirement that a GEMM 2 facility must comply with in a calendar year as determined pursuant to Part B, Section I.A.
- II.EE. “GHG best available emission control technology” (GHG BAECT) means a GHG emission control technology for a GHG emission unit based on the maximum degree of GHG reductions achievable on a case-by-case basis, taking into account energy, environmental, and economic impacts, employment of which is demonstrated by compliance with the GHG BAECT and energy BMP intensity rate determination.

- II.FF. “GHG BAECT and energy BMP intensity rate” means the total direct GHG emissions per unit of production from the emissions units within the audit scope after GHG BAECT and energy BMPs are operational as determined in Part C.
- II.GG. “GHG credit” means a uniquely identifiable and tradable compliance instrument equal to one metric ton of CO₂e reduced, which is generated, issued, transferable, and may be retired pursuant to Part D. The GHG credit must be real, additional, quantifiable, permanent, verifiable, and enforceable and provide additional emissions reductions beyond a facility’s compliance obligation.
- II.HH. “GHG crediting and tracking system” means a GHG credit accounting, tracking, and trading system established by the Division and/or its agent where GHG credits are issued by the Division or its agents to manufacturing stationary sources and midstream segment companies in the system, and may be transferred between regulated sources, and retired under this Regulation Number 27 and Regulation Number 7.
- II.II. “GHG mitigation plan” means the plan produced by a GEMM 2 facility pursuant to Part A, Section III.B.1.
- II.JJ. “GHG reduction plan” means the plan produced by a GEMM 2 facility under Part B, Section II.A., or the plan produced by a stationary source constructed on or before the effective date of this rule that becomes a GEMM 2 facility after the effective date of this rule under Part B, Section I.B, as applicable.
- II.KK. “Greenhouse gas” (GHG) means carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆) and Nitrogen Trifluoride (NF₃).
- II.LL. “Harmful air pollutant” as used in this section means pollutants designated by EPA as criteria air pollutants (carbon monoxide, lead, nitrogen dioxide, ozone, particulate pollution (PM) (PM_{2.5} and PM₁₀) and sulfur dioxide) or hazardous air pollutants. This term is not intended to articulate different thresholds or standards for those pollutants listed as harmful air pollutants than currently established under the federal Clean Air Act or the Colorado Air Pollution Prevention and Control Act and their respective implementing regulations.
- II.MM. “Independent third party” means an engineering or consulting firm selected by the State of Colorado; which is not affiliated with the stationary source, its subsidiaries, or related entities and has no common ownership with the stationary source. The capabilities and knowledge of the firm must include, but are not limited to, background, experience, and recognized abilities to perform the assessment activities, data analysis, and report preparation and experience working with the industry subject to this section.
- II.NN. “International Organization of Standardization” (ISO) means the independent, non-governmental international standard-setting body composed of representatives from various national standards organizations.
- II.OO. “ISO 50001: Energy Management Systems – Requirements with guidance for use” (ISO 50001) means the internationally accepted standard which specifies the requirements for an organization to demonstrate that it has a sustainable energy management system in place, has completed the energy planning process, and has a commitment to continual improvement of its energy performance.
- II.PP. “Lead auditor” means an individual who has met the requirements of and is certified as a lead auditor through a professional certification body.

- II.QQ. "Management system" means the policies, processes, and procedures used by an organization to ensure that it can fulfill the tasks required to achieve its GHG emissions or energy management objectives.
- II.RR. "Manufacturing stationary source" means an EITE stationary source or a GEMM 2 facility.
- II.SS. "Midstream segment company" means "midstream company" as defined in Regulation Number 7, Part B, Section VII.
- II.TT. "Municipal gas distribution utility" means a municipally owned utility that provides gas service to more than ninety thousand customers.
- II.UU. "Net meter" means a renewable energy resource or renewable energy storage on the EITE stationary source's property which supply energy directly to the EITE stationary source's energy provider in exchange for a Power Purchase Agreement where the customer receives credit for the energy production.
- II.VV. "Non-GHG BAECT emissions" means the GHG mass emissions from an EITE stationary source that are not covered by the audit.
- II.WW. "North American industry classification system (NAICS) code(s)" means the six-digit code(s) that represents the product(s)/activity(s)/service(s) at a facility or supplier as listed in the Federal Register and defined in "North American Industrial Classification System Manual 2007," available from the U.S. Department of Commerce, National Technical Information Service, Alexandria, VA 22312 (as published March 13, 2023).
- II.XX. "Permanent" means the GHG emission reductions are not reversible, or, when the GHG emission reductions are reversible, that mechanisms are in place to replace any reversed emission reductions to ensure that all reductions that are awarded GHG credits endure.
- II.YY. "Plain-language" means writing that is clear, concise, well-organized, and follows other best practices appropriate to the subject or field and is easily understandable.
- II.ZZ. "Primary account representative" means an individual authorized by a manufacturing stationary source to make submissions to the Division or its agent in all matters pertaining to this Regulation Number 27 that legally bind the authorizing source.
- II.AAA. "Process" means a specific operation at an EITE stationary source comprising a series of actions or steps which are carried out in a specific order to complete a particular stage in the manufacturing process.
- II.BBB. "Product" means the quantifiable material output of an individual manufacturing process or manufacturing facility.
- II.CCC. "Proof of certification" means an official document issued by the formal registrar or certifying body stating the scope of certification, the expiration date and the standards to which the stationary source is certified.

- II.DDD. "Qualified third-party auditor" means one or more individuals who hold a valid lead auditor certification in greenhouse gas and/or energy management systems and have demonstrated capabilities to evaluate GHG reduction opportunities for large, energy-intensive, industrial manufacturing processes and facilities. Qualified third-party auditors must have worked as an auditor for at least two years, or must have worked as a project manager or lead person for not less than four years (two of which may be graduate level work) in: (1) the development of GHG or other air emission inventories, or (2) as a lead environmental data or financial auditor. The auditor must not be affiliated with the EITE stationary source, its subsidiaries, or related entities; there can be no common ownership between the EITE stationary source and the third-party auditor. The capabilities and knowledge of the auditor must include, but are not limited to, background, experience, and recognized abilities to perform the assessment activities, data analysis, and report preparation and experience lead auditing GHG or energy management systems for the industry subject to this section.
- II.EEE. "RACT/BACT/LAER Clearinghouse" (RBLC) means EPA's central database of air pollution technology information, including past RACT, BACT, and LAER decisions contained in New Source Review (NSR) permits, to promote the sharing of information among permitting agencies and to aid in future case-by-case determinations.
- II.FFF. "Real" means that GHG emission reductions result from a demonstrable action or set of actions and are quantified using appropriate, accurate, and conservative methodologies.
- II.GGG. "Recovered methane project owner or operator" means a owner or operator of a recovered methane project that is eligible to generate recovered methane credits under Regulation Number 22, Part C.
- II.HHH. "Regulated source" means a source of greenhouse gas that is subject to a rule adopted by the Commission under Section 25-7-105(1)(e), C.R.S., that imposes specific and quantifiable GHG reduction obligations upon that source or group of sources.
- II.III. "Residential building unit" means a building or structure designed for use as a place of residency by a person, a family, or families. The term includes manufactured, mobile, and modular homes, except to the extent that any such manufactured, mobile, or modular home is intended for temporary occupancy or for business purposes. Each individual residence within a building will be counted as one Residential building unit.
- II.JJJ. "Residential community" means an area where more than ten (10) residential building units are grouped together within a one (1) mile radius.
- II.KKK. "Retail distributed generation" means a renewable energy resource or renewable energy storage that directly supplies building or process energy needs at a metered location, where surplus energy is supplied to the location's energy provider when energy production is greater than on-site demand and grid energy is supplied through a customer meter to the location during times when on-site production is less than demand.
- II.LLL. "Small gas distribution utility" means a public utility providing gas service to ninety thousand retail customers or fewer. Small gas distribution utility does not include a municipal gas distribution utility.

- II.MMM. “Social cost of GHGs” means the monetized damages associated with an incremental increase in GHG emissions in a given year. The social cost of GHGs must include separate calculations for carbon, methane, and nitrous oxide, and the social cost of any other GHGs must be calculated using carbon dioxide equivalent. For purposes of Regulation Number 27, the social cost of GHGs is established, using a two and one-half percent discount rate, by the Federal Interagency Working Group’s Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide, Interim Estimates under Executive Order 13990, dated February 2021.
- II.NNN. “Strategic energy management” (SEM) means a management system-based, continuous improvement approach to energy management that seeks to improve an organization’s energy performance, reduce energy costs, and reduce GHG emissions associated with energy use; drives improvement in facility energy efficiency through equipment upgrades operations and maintenance improvements and behavioral changes.
- II.OOO. “Technically feasible” means that the GHG reduction measure can be implemented at the facility within existing technological and scientific limitations.
- II.PPP. “Transfer” means, as to a GHG credit, the removal of a GHG credit from one manufacturing stationary source’s account in the GHG crediting and tracking system and placement of that credit into another manufacturing stationary source’s account in the system if agreed to by both the transferor and transferee manufacturing stationary sources.
- II.QQQ. “Verifiable” means that a reported emission reduction resulting from a GHG credit at a regulated source is well documented and transparent such that it lends itself to an objective review by the Division to verify the emission reduction is real, using monitoring of emissions reductions relative to the GEMM 2 annual GHG emissions requirement or annual emissions limitation, as applicable, for the relevant compliance year.

III. Noncompliance

- III.A. In the event that an EITE stationary source fails to meet its annual emissions limitation, the EITE stationary source will be deemed in noncompliance and must surrender or obtain and surrender three (3) GHG credits for every metric ton of CO₂e emitted by the EITE stationary source in excess of the annual emissions limitation and may be subject to a civil penalty or other enforcement action by the Division.
- III.B. In the event that a GEMM 2 facility fails to comply with: (i) its aggregated GHG emissions requirement for the first compliance period (calculated by aggregating a facility’s GEMM 2 annual GHG emission requirements for 2024, 2025, and 2026); (ii) its aggregated GHG emissions requirement for the second compliance period (calculated by aggregating a facility’s GEMM 2 annual GHG emission requirements for 2027, 2028, and 2029); or (iii) after 2029, its GEMM 2 annual GHG emissions requirement in any compliance year, the facility’s GEMM 2 annual GHG emissions requirement will be adjusted downwards by at least two (2) times the amount, in metric tons of CO₂e, by which the facility exceeded its aggregated GHG emissions requirements in either compliance period or, after 2029, its GEMM 2 annual GHG emissions requirement. The timeline required for the GEMM 2 facility to achieve the mitigation will be determined by the Division but must be no later than three (3) years after the period or year of non-compliance.
- III.B.1. If a GEMM 2 facility fails to comply with its aggregated GHG emissions requirements in either compliance period or, after 2029, its GEMM 2 annual GHG emissions requirement in any compliance year, the facility must submit a GHG mitigation plan for Division review and approval by no later than December 31 of the year following the period or year of non-compliance, documenting how it will comply with the additional required GHG reductions. The GHG mitigation plan must include:

- III.B.1.a. The amount by which the GEMM 2 facility exceeded the facility's aggregated GHG emissions requirement in metric tons of CO₂e for the previous period or GEMM 2 annual GHG emissions requirement in metric tons of CO₂e for the previous year.
- III.B.1.b. The adjusted GHG emissions requirement for the mitigation period, as determined by the Division pursuant to Section III.B.
- III.B.1.c. The onsite GHG reduction measure(s) being implemented to achieve the additional reduction required pursuant to Section III.B, for the mitigation period.
- III.C. If the GEMM 2 facility does not achieve the applicable GEMM 2 annual GHG emissions requirement or mitigation reduction amount, the facility must take the necessary corrective actions to achieve the required reductions pursuant to this Section III, in addition to paying any civil penalties that may be assessed. The GEMM 2 facility may be deemed out of compliance for the entirety of the year and subject to daily civil penalties.
- III.D. In the event that a GEMM 2 facility fails to implement the emission reduction measures as documented in the Division-approved GHG reduction plan pursuant to Part B, Section II, the GEMM 2 facility may be deemed out of compliance and may be subject to enforcement action and daily civil penalties.
- III.E. If a GEMM 2 facility does not submit timely, complete, and accurate documentation to the Division pursuant to any section of Regulation Number 27, the facility may be subject to enforcement action including assessment of daily civil penalties.
- III.F. Nothing in this Section III limits the enforcement powers of the Division under the Act to remedy noncompliance with Regulation Number 27, including but not limited to the Division's ability to seek additional penalties, and compel actual reductions at any manufacturing source in noncompliance.

PART B GEMM 2 Facility Requirements

I. Greenhouse Gas Emissions Reduction Requirements for GEMM 2 Facilities

- I.A. Stationary sources qualifying as GEMM 2 facilities as of January 1, 2023 must comply with the GEMM 2 annual GHG emissions requirements set forth in Part B, Section I.A., as applicable. An EITE stationary source that emitted equal to or greater than 25,000 mt of CO₂e emissions as of January 1, 2023, but was not subject to Part C of this Regulation Number 27 as of January 1, 2023, is subject to the GEMM 2 annual GHG emissions requirements set forth in Part B, Section I.A., as applicable, unless the source notifies the Division within 6 months of the effective date that it will comply with Part C. Compliance will be demonstrated pursuant to Part B, Section IV.
 - I.A.1. The owner or operator of a GEMM 2 facility for which the higher of the GEMM 2 facility's 2021 or 2022 emissions, as reported pursuant to Regulation Number 22, Part A, reflect a reduction in direct GHG emissions of at least twenty (20) percent, as compared to the facility's 2015 GHG emissions, must comply with the following requirements:
 - I.A.1.a. The owner or operator of the GEMM 2 facility must comply with the GEMM 2 annual GHG emissions requirements in Table 1.

Table 1

Year	GEMM 2 Annual GHG Emissions Requirement
2024 – 2029	GEMM 2 facility GHG baseline emissions
2030 and beyond	1% less than the GEMM 2 facility GHG baseline emissions

I.A.1.b. The owner or operator of a GEMM 2 facility subject to Section I.A.1.a., above, may, in the alternative to complying with Section I.A.1.a., comply with the following requirements.

I.A.1.b.(i) For calendar years 2024 and 2025, comply with a GEMM 2 annual GHG emissions requirement equal to seventy-five (75) percent of the facility's 2015 reported emissions;

I.A.1.b.(ii) Beginning in 2026, comply with a GEMM 2 annual GHG emissions requirement equal to the facility's GEMM 2 facility GHG baseline emissions; and

I.A.1.b.(iii) Beginning in 2030 and continuing in perpetuity, comply with a GEMM 2 annual GHG emissions requirement for 2030, as established under Section I.A.1.a, above, subject to any additional reductions required under Section I.A.5.; provided, however, that

I.A.1.b.(iv) If the GEMM 2 facility subject to this Section I.A.1.b. cannot meet a GEMM 2 annual GHG emissions requirement equal to its GEMM 2 facility GHG baseline emissions by 2026 consistent with Section I.A.1.b.(ii), it must either:

I.A.1.b.(iv)(A) Beginning in 2028 and continuing in perpetuity, comply with its GEMM 2 annual GHG emissions requirement for 2030, as established under Section I.A.1.b.(iii), subject to an additional one (1) percent reduction; or

I.A.1.b.(iv)(B) Beginning in 2030 and continuing in perpetuity, comply with its GEMM 2 annual GHG emissions requirement for 2030, as established under Section I.A.1.b.(iii), subject to an additional two (2) percent reduction.

I.A.2. The owner or operator of a GEMM 2 facility for which the higher of the GEMM 2 facility's 2021 or 2022 emissions, as reported pursuant to Regulation Number 22, Part A reflect a reduction in direct GHG emissions of at least ten (10) percent but less than twenty (20) percent, as compared to the facility's 2015 GHG emissions, must comply with the GEMM 2 annual GHG emissions requirements in Table 2.

Table 2

Year	GEMM 2 Annual GHG Emissions Requirement
2024 – 2029	1.25% less than the GEMM 2 facility GHG baseline emissions
2030 and beyond	7% less than the GEMM 2 facility GHG baseline emissions

- I.A.3. The owner or operator of a GEMM 2 facility for which the higher of the GEMM 2 facility's 2021 or 2022 emissions, as reported pursuant to Regulation Number 22, Part A reflect a reduction in direct CO₂e emissions greater than zero (0) percent but less than ten (10) percent, as compared to its 2015 GHG emissions, must comply with the GEMM 2 annual GHG emissions requirements in Table 3.

Table 3

Year	GEMM 2 Annual GHG Emissions Requirement
2024 – 2029	1.50% less than the GEMM 2 facility GHG baseline emissions
2030 and beyond	8% less than the GEMM 2 facility GHG baseline emissions

- I.A.4. The owner or operator of a GEMM 2 facility for which the higher of the GEMM 2 facility's 2021 or 2022 emissions, as reported pursuant to Regulation Number 22, Part A are equal to or greater than its 2015 GHG emissions, must comply with the GEMM 2 annual GHG emissions requirements in Table 4.

Table 4

Year	GEMM 2 Annual GHG Emissions Requirement
2024 – 2029	1.75% less than the GEMM 2 facility GHG baseline emissions
2030 and beyond	12.5% less than the GEMM 2 facility GHG baseline emissions

- I.A.5. Beginning in 2030, and for each year thereafter, in addition to those limits in Part B, Sections I.A.1. through I.A.4., GEMM 2 facilities will also be required to incorporate the reductions in Table 5 into the GEMM 2 annual GHG emissions requirement in 2030 and beyond, based on the GEMM 2 facility's percent contribution towards the cumulative GEMM 2 facilities' higher of 2021 or 2022 emissions, as reported pursuant to Regulation Number 22, Part A.

Table 5

GEMM 2 Facility Percent Contribution	GEMM 2 Annual GHG Emissions Requirement in 2030 and beyond
30% or greater	6% less than the GEMM 2 facility GHG baseline emissions
At least 20% but less than 30%	5% less than the GEMM 2 facility GHG baseline emissions
At least 10% but less than 20%	4% less than the GEMM 2 facility GHG baseline emissions
At least 5% but less than 10%	3% less than the GEMM 2 facility GHG baseline emissions

- I.A.6. If the owner or operator of a GEMM 2 facility certifies through a Division-approved form that it will voluntarily comply with the facility's GEMM 2 annual GHG emissions requirement for 2030 beginning no later than 2025 and each year thereafter through limiting onsite direct GHG emissions, the facility is not subject to the requirements in Part B, Section II. Upon submission of this form, the facility's GEMM 2 annual GHG emissions requirement for 2030 is established as the facility's new, enforceable GEMM 2 annual GHG emissions requirement for calendar year 2025 and each year thereafter. To qualify, the owner or operator of the GEMM 2 facility must provide the Division-approved form, certified by a responsible party, to the Division no later than May 31, 2025.
- I.B. Except as otherwise provided in this Section I.B, any stationary source constructed on or before the effective date of this rule that becomes a GEMM 2 facility after the effective date of this rule must comply with the following requirements. An EITE stationary source that emitted equal to or greater than 25,000 mt of CO₂e emissions as of January 1, 2023, but was not subject to Part C of this Regulation Number 27 as of January 1, 2023, may not elect to comply with this Regulation Number 27 through the requirements of Part B, Section I.B.
- I.B.1. By December 31st of the year in which the facility first reports direct GHG emissions equal to or greater than 25,000 metric tons of CO₂e, the owner or operator of the facility must conduct a GHG emission control audit to assess the implementation of GHG best available emission control technology (GHG BAECT), propose a five (5) year GHG reduction plan for the facility and submit the audit report and GHG reduction plan to the Division. An EITE stationary source's GHG reduction plan for purposes of this Section I.B. must propose to achieve greater than a five (5) percent reduction in direct GHG emissions from the EITE stationary source's current direct GHG emissions, compared to the most recent year of operation. The GHG BAECT analysis must be performed according to the requirements set out in Part C, Sections I.C.1.a.
- I.B.2. Audits must be conducted by a qualified third-party auditor and meet or exceed nationally or internationally accepted energy and GHG accounting and management audit standards or protocols.
- I.B.3. The Division will post the facility's audit report and GHG reduction plan proposal to the Division's website within fifteen (15) days of receipt and provide a 30-day public comment period.

- I.B.4. The Division will review the audit report, GHG reduction plan and public comments received and issue a final five (5) year GHG reduction plan for the facility within one hundred twenty (120) days of receipt of the report and plan. The Division will post the facility's approved GHG reduction plan on the Division's website.

II. Greenhouse Gas Reduction Plans

- II.A. No later than September 30, 2025, except as otherwise provided in this Regulation Number 27, the owner or operator of a GEMM 2 facility subject to Section I.A. must develop a GHG reduction plan and submit the certified GHG reduction plan on a Division-approved form, along with the independent third-party evaluation and findings of the plan, to the Division. The owner or operator of a glass container manufacturing facility must submit the certified GHG reduction plan, as described in this Section II. as expeditiously as practicable but no later than June 1, 2027. The GHG reduction plan must include the following information:

II.A.1. Basic emissions information

II.A.1.a. The GEMM 2 facility's GHG baseline emissions.

II.A.1.b. The difference in GHG emissions in metric tons of CO₂e reported for the year 2024, compared to its GEMM 2 facility GHG baseline emissions.

II.A.1.c. The percent of GHG emission reduction required for the GEMM 2 facility to comply with its 2030 GHG emissions requirement pursuant to Section I.A. based on the GEMM 2 facility's GHG baseline emissions, as applicable.

I.A.1.d. The difference in metric tons of CO₂e between the GEMM 2 facility's GEMM 2 annual GHG emissions requirement for 2030 and its GEMM 2 annual GHG emissions requirement for 2024, as applicable. This should be calculated as:

GEMM 2 annual GHG Emissions Requirement for 2030 - GEMM 2 annual GHG emissions requirement for 2024

II.A.2. GHG reduction measures, including portfolio approach

II.A.2.a. The list of all GHG reduction measures that result in greater than de minimis GHG reductions and that are technically feasible and commercially available or other measures that facilities propose for implementation at the GEMM 2 facility. The following information is required for each measure listed.

II.A.2.a.(i) The quantity of metric tons of CO₂e reduced per year from each measure;

II.A.2.a.(ii) The net reduction in associated harmful air pollution in metric tons per year from each measure;

II.A.2.a.(iii) The cost in USD per metric ton of CO₂e reduced for each reduction measure for all measures;

II.A.2.a.(iv) The timeframe for implementation for each measure; and

II.A.2.a.(v) If a facility seeks to utilize Section II.A.3.b., below, then information as to those GHG reduction measures that are alternate options or redundant or incompatible with each other.

- II.A.3. The portfolio of measures up to the 2030 social cost of GHGs that the facility is required to propose to implement by 2030 towards achievement of the facility's 2030 GHG emissions requirement as well as any measure(s) above the 2030 social cost of GHGs, including the cost information and estimated reduction of harmful air pollution of those measure(s), that the facility is voluntarily proposing to implement by 2030 to ensure it achieves its 2030 GEMM 2 GHG emissions requirement.
 - II.A.3.a. If the facility proposes to implement a portfolio measures from its GHG reduction plan that is estimated to yield total GHG reductions within five (5) percent of another portfolio of measures, the facility must propose to implement the portfolio of measures that provides the greater reduction in harmful air pollution, provided that the cost of the portfolio of measures is at or below the 2030 social cost of GHGs.
 - II.A.3.b. A facility is not required to select and implement measures that are alternate options or redundant or incompatible with each other.
- II.A.4. If, as of the GEMM 2 facility's submittal deadline for its GHG reduction plan in this Section II.A., a facility is already in the process of constructing or implementing, including post-construction project implementation or ramp up, a GHG reduction measure or portfolio of measures that are projected to achieve the entirety of the facility's 2030 GEMM 2 annual GHG emissions requirement, the facility's list of GHG reduction measures in Section II.A.2. need only include such measure(s), and the facility must only propose to implement such measure(s) to comply with Section II.A.3. A facility that qualifies for this section is not subject to Section II.A.3.a.
- II.A.5. If the GEMM 2 facility proposes to implement all technically feasible portfolio of measures at or below the 2030 social cost of GHGs, but the proposed measures do not satisfy the 2030 GEMM 2 GHG emissions requirement, the GHG reduction plan may indicate that the facility plans to use the GHG crediting and tracking system for compliance, provided that the GEMM 2 facility complies with Section II.A.6. if it is located within one (1) mile of a disproportionately impacted community and within fifteen (15) miles of a residential community.
- II.A.6. If (1) a GEMM 2 facility plans to use the GHG crediting and tracking system for compliance and (2) any portion of the GEMM 2 facility's property line is within one (1) mile of a disproportionately impacted community and within fifteen (15) miles of a residential community as of the effective date of this rule, the following requirements apply:
 - II.A.6.a. The GHG reduction plan must identify additional GHG reduction measures not yet selected for implementation under Part B, Sections II.A.3. that cost up to fifty (50) percent above the 2030 social cost of GHGs. If any such measures exist, the GHG reduction plan must quantify the largest amount of harmful air pollution reduction achievable by implementing one or a combination of such measures, in each case up to the facility's GEMM 2 annual GHG emissions requirement for 2030.
 - II.A.6.b. The GHG reduction plan must propose to implement measure(s) to reduce harmful air pollution in an amount equal to the amount achievable identified in Section II.A.6.a.
- II.B. The owner or operator of the GEMM 2 facility must ensure one of its responsible agents certifies that the contents of the GHG reduction plan documentation is complete and accurate.

- II.C. The owner or operator of the GEMM 2 facility must ensure an independent third party conducts a technical and regulatory review of its GHG reduction plan. The selected firm will review the GHG reduction plan to determine the accuracy and completeness of the plan including, without limitation, cost projections, assumptions and data sources, GHG emission and harmful air pollution impacts, and compliance with this Part B, Section II.
- II.D. The GEMM 2 facility must cooperate with the independent third party to assure accuracy and completeness of the GHG reduction plan, and compliance with this Part B, Section II. Upon completion of the review, the plan must be certified by the independent third party as adhering to the requirements of this Part B, Section II, prior to submission of the plan to the Division.
- II.E. The GHG reduction plan must include a plain-language summary of the proposed GHG reduction plan for the GEMM 2 facility.
 - II.E.1. Each plain language summary shall be provided in the top two languages spoken by the community surrounding the GEMM 2 facility. The translations must be prepared by a certified translator.
 - II.E.2. The plain language summary shall include the list of measures analyzed in the plan for the GEMM 2 facility, the correlating GHG emission and harmful air pollutant impacts of the measures, and justification for why the proposed measures were identified.
- II.F. Within fifteen (15) days of receipt, the Division will post the certified GHG reduction plan and the independent third party review of the GHG reduction plan on the Division's website.
- II.G. The Division will hold a thirty (30) day public comment period for GHG reduction plans.
- II.H. Except for GHG reduction plans submitted by glass container manufacturing facilities after September 30, 2025 but no later than June 1, 2027 under Part B, Section II.A, the Division must issue an approval or request for modification of the proposed GHG reduction plan to the GEMM 2 facility by December 31, 2025. For GHG reduction plans submitted by glass container manufacturing facilities after September 30, 2025 but no later than June 1, 2027, the Division must issue an approval or request for modification of the proposed GHG reduction plan to the GEMM 2 facility within three months of submission.
- II.I. The Division will hold at least three (3) public meetings to review the approved GEMM 2 facility GHG reduction plans.
- II.J. GHG reduction measures must be timely and completely implemented in accordance with the GEMM 2 facility's documented and approved GHG reduction plan. A GEMM 2 facility may request to modify the GHG reduction plan for the facility at any time. The modification must comply with the same requirements for the GHG reduction plan in Part B, Section II.
- III. GEMM 2 Facility Greenhouse Gas Emission Requirement Compliance**
- III.A. A GEMM 2 facility must first attempt to meet its GEMM 2 annual GHG reduction requirement through technically feasible, onsite measures at or below the 2030 social cost of GHGs.
 - III.A.1. GHG reductions from onsite carbon capture and storage are considered onsite measures for purposes of Section III.A; provided, however, that onsite carbon capture and storage reductions may only be counted towards compliance with this Regulation Number 27 if the Division has established or adopted by reference a standardized carbon capture and storage protocol or protocols.

- III.B. If a facility cannot meet its GEMM 2 annual GHG reduction requirement through technically feasible, onsite measures at or below the 2030 social cost of GHGs, the facility may retire GHG credits through the GHG crediting and tracking system, pursuant to Part D, to achieve the remainder of its GEMM 2 annual GHG emission requirement, provided the facility complies with requirements set forth in Part B, Section II.A.6., as applicable.
- III.C. Permitting requirements for GHG emission reduction measures.
- III.C.1. A manufacturing stationary source that requires a construction permit(s) or modification(s) of an existing permit(s) in accordance with Regulation Number 3 to comply with the requirements of Parts B or C of Regulation Number 27 must submit a complete permit application to the Division at least twelve (12) months prior to the start of construction or at least twelve (12) months prior to the start of the modification.
- III.C.2. If the owner or operator of a GEMM 2 facility has complied with Part B, Section III.C.1. and the Division has not issued the permit required to comply with Part B within twelve (12) months of receipt of the permit application, the Division will adjust the facility's GEMM 2 annual GHG emissions requirement, as applicable, upon submission of documentation by the GEMM 2 facility that shows good cause.

IV. Reporting and Annual Compliance Certification Requirements

- IV.A. Beginning March 31, 2025 and every March 31 thereafter, owners or operators of GEMM 2 facilities must submit an initial report to the Division that provides the GEMM 2 facility's reported direct GHG emissions for the previous year and the difference between such reported emissions and the GEMM 2 facility's emissions requirement for the previous year, as determined pursuant to Part B, Section I.A. Such report must include the following:
- IV.A.1. For a GEMM 2 facility utilizing a combined heat and power unit, when calculating the GEMM 2 facility's annual GHG emissions in metric tons of CO₂ pursuant to Section IV.A.1.a., the facility may account for the GHG emissions reduction associated with the combined heat and power unit's displaced direct thermal emissions, using a six-step formula submitted pursuant to a Division approved form. GHG emissions reductions resulting from the utilization of a facility's combined heat and power unit(s) may only account for up to 50% of the facility's GEMM 2 annual GHG emissions reduction requirement pursuant to Section I.A.1. through Section I.A.5. of this Part.
- IV.A.2. The GEMM 2 annual GHG emissions requirement for the previous year, as determined pursuant to Part B, Section I.A.
- IV.A.3. The difference, if any, between the GEMM 2 facility's reported direct GHG emissions for the previous year and the GEMM 2 facility's GEMM 2 annual GHG emissions requirement for the previous year.
- IV.B. On September 30, 2027, owners or operators of GEMM 2 facilities must submit a report to the Division that demonstrates compliance on a three-year cycle and includes:
- IV.B.1. A compliance certification in which the account representative of each GEMM 2 facility certifies:
- IV.B.1.a. The GEMM 2 facility's GHG emissions, in metric tons of CO₂e, for the calendar years 2024, 2025, and 2026.
- IV.B.1.b. The GEMM 2 facility's GHG emissions requirement for the calendar years 2024, 2025, and 2026, as determined pursuant to Part B, Section I.A.

- IV.B.1.c. The difference, if any between the GEMM 2 facility's total reported direct GHG emissions for calendar years 2024, 2025, and 2026 and the GEMM 2 facility's total GEMM 2 annual GHG emissions requirements for 2024, 2025, and 2026 combined.
 - IV.B.1.d. Documentation of GHG credits retired prior to expiration in the GHG crediting and tracking system for compliance purposes in the applicable period.
 - IV.B.2. A report detailing progress of and compliance with the approved GHG reduction plan for the GEMM 2 facility including relevant project management documentation, project status, and timeline
- IV.C. On September 30, 2030, owners or operators of GEMM 2 facilities must submit a report to the Division that demonstrates compliance on a three-year cycle that includes:
 - IV.C.1. A compliance certification in which the account representative of each GEMM 2 facility certifies:
 - IV.C.1.a. The GEMM 2 facility's GHG emissions, in metric tons of CO₂e, for the calendar years 2027, 2028, and 2029.
 - IV.C.1.b. The GEMM 2 facility's GHG emissions requirement for the calendar years 2027, 2028, and 2029, as determined pursuant to Part B, Section I.A.
 - IV.C.1.c. The difference, if any between the GEMM 2 facility's total reported direct GHG emissions for calendar years 2027, 2028, and 2029 and the GEMM 2 facility's total GEMM 2 annual GHG emissions requirements for 2027, 2028, and 2029 combined.
 - IV.C.1.d. Documentation of GHG credits retired prior to expiration in the GHG crediting and tracking system for compliance purposes in the applicable period.
 - IV.C.2. A report detailing progress of and compliance with the approved GHG reduction plan for the GEMM 2 facility including relevant project management documentation, project status, and timeline.
- IV.D. Beginning September 30, 2031 and every September 30 thereafter, owners or operators of GEMM 2 facilities must submit an annual report to the Division that includes:
 - IV.D.1. A compliance certification in which the account representative of each GEMM 2 facility certifies:
 - IV.D.1.a. The GEMM 2 facility's annual GHG emissions, in metric tons of CO₂e, for the previous calendar year.
 - IV.D.1.b. The GEMM 2 annual GHG emissions requirement for the previous year, as determined pursuant to Part B, Section I.A.
 - IV.D.1.c. The difference, if any, between the GEMM 2 facility's reported direct GHG emissions for the previous year and the GEMM 2 facility's GEMM 2 annual GHG emissions requirement for the previous year.
 - IV.D.1.d. Documentation of GHG credits retired prior to expiration in the GHG crediting and tracking system for compliance purposes in the applicable year.

- IV.D.2. A report detailing progress of and compliance with the approved GHG reduction plan for the GEMM 2 facility including relevant project management documentation, project status, and timeline.
- IV.E. If (1) Regulation Number 22, Part A is updated such that additional emission sources are identified that are required to be reported from a GEMM 2 facility, which emission sources existed prior to the change in required reporting under Regulation Number 22, Part A, (2) methodologies for emissions calculations change; or (3) there is discovery of a reporting error, then the Division will consult with the GEMM 2 facility to adjust the GEMM 2 facility's previous years' reported GHG emissions totals to accommodate such changes or to otherwise ensure a consistent approach to a facility's emissions calculations, back to the year 2015, as applicable.
- V. Recordkeeping**
- V.A. GEMM 2 facilities must maintain records for a period of ten (10) years and make records available to the Division upon request.
- V.A.1. Reports required under Part B, Section IV.
- V.A.2. Division-approved GHG reduction plans submitted pursuant to Part B, Section I.B. or Section II.A. as applicable, and records reasonably necessary to demonstrate compliance with the approved GHG reduction plan.
- V.A.3. Any other documents submitted to the Division under this Regulation Number 27.
- V.B. Within fifteen (15) days of receipt, the Division will post to its website the following documents that may be submitted by a GEMM 2 facility under this Regulation Number 27: certifications submitted under Part B, Section I.A.6; reports submitted under Part B, Section IV.A; records demonstrating compliance with GHG reduction plans submitted under Part B, Section V.A.2; and registration applications submitted under Part D, Section II.A.1.
- V.C. Confidential business information contained in records submitted to the Division by GEMM 2 facilities under this Regulation Number 27 must be clearly identified and be submitted in a separate, supplementary document to the records.

PART C Energy-Intensive Trade-Exposed Stationary Source Requirements

I. Audit Requirements

- I.A. Energy and GHG Emission Control Audits
- I.A.1. By December 31, 2022, and December 31 every five years thereafter, owners or operators of each EITE stationary source must conduct energy and GHG emission control audits to establish greenhouse gas best available emission control technology (GHG BAECT) and energy best management practices (energy BMPs) and determine whether the stationary source is employing GHG BAECT and energy BMPs at the EITE stationary source, and submit the audit report to the Division.
- I.A.2. Each EITE stationary source must conduct an audit within twelve (12) months of reporting direct GHG emissions equal to or greater than 25,000 metrics tons CO₂e per year under Regulation Number 22, Part A and/or 40 CFR Part 98 and every five years thereafter.

- I.A.3. Audits must be conducted by a qualified third-party auditor and meet or exceed nationally or internationally accepted energy and GHG accounting and management audit standards or protocols.

I.B. Audit Plan

- I.B.1. Each EITE stationary source must submit an audit plan to the Division for approval at least 120 days prior to beginning the audit as required in Section I.A. The Division will review the audit plan and notify the EITE stationary source within 60 days of submission of any deficiencies. If notified of deficiencies, the EITE stationary source must submit a revised audit plan for final approval no later than 30 days prior to beginning the audit. The EITE stationary source must receive approval from the Division of the audit plan prior to beginning the audit. Such approval shall not be unreasonably withheld. The audit plan must include:
 - I.B.1.a. A description of all the emission units at the EITE stationary source that directly release one or more GHGs, ordered from largest emitting to smallest emitting emission units averaged over the past 5 years, quantified in CO₂e as well as broken out by GHG type. Using this list, the emission units that comprise at least the top 80% of the EITE stationary source's direct GHG emissions shall be identified, documented and included in the GHG emissions audit scope. Additionally, any individual emissions unit that comprises 2% or more of the EITE stationary source's direct GHG emissions shall be identified, documented and included in the GHG emissions audit scope.
 - I.B.1.b. Unless the EITE stationary source is utilizing Section I.B.1.c. to show employment of energy BMPs, a description of all the emission units at the EITE stationary source that consume energy, averaged over the past five (5) years and ordered from largest consuming to smallest consuming. Using this list, the energy consumption sources that comprise the top 80% of the EITE stationary source's energy consumption shall be identified, documented and included in the energy audit scope.
 - I.B.1.c. If an EITE stationary source is certified to the Federal Energy Star Program or ISO 50001, the EITE stationary source will be determined to be employing energy BMPs and the energy audit will be limited in scope. Proof of certification of the EITE stationary source to one or more of these existing standards must be included in the audit plan to utilize this option.
 - I.B.1.c.(i) If an EITE stationary source is planning on becoming certified to one of these standards, the EITE stationary source must submit the certification or registration timeline or plan, including key milestones towards certification or registration with certification scheduled for no more than 12 months after the audit date.
 - I.B.1.d. Records of any previous third-party audit results that the EITE stationary source proposes to use to support the audit on a supplementary basis or to avoid duplication of data collection efforts that have been performed within three (3) years prior to the planned audit date. To be accepted, supplementary audit data must be verified and validated by a third party and result from an audit that meets or exceeds nationally or internationally accepted energy and GHG accounting audit standards or protocols.

I.B.1.e. A description of the audit team members, including experience, qualifications, and role in the audit, and existing or previous business relationship, and the nature of such relationship with the owner or operator of the EITE stationary source. If there is an existing or previous business relationship, a list and description of work done for the owner or operator of the EITE stationary source.

I.B.1.e.(i) The Division may reject a proposed qualified third-party auditor or audit team if it does not meet the qualifications in Sections II.H. and II.KK., failed to conduct a previous audit to the satisfaction of the Division, or is deemed to have a previous or existing relationship with the source that is so pervasive that the auditor would be unable to conduct the audit in an unbiased and independent manner.

I.B.1.f. The specific GHG and/or energy audit standards, protocols or procedures to be used for conducting the audit, if applicable.

I.C. Audit Reports

I.C.1. Each EITE stationary source must complete the audit report in accordance with Sections I.C.1.a. through III.C.1.e. and submit the audit report to the Division by December 31 of the audit year that includes the following elements for all GHG emission units listed in accordance with Section I.B. and specified in the Division-approved audit plan, at a minimum.

I.C.1.a. The GHG BAECT analysis. The audit team must analyze GHG BAECT as follows:

I.C.1.a.(i) Identify all available control technologies and strategies with practical potential for application to reduce GHG emissions from the GHG emission units included in the audit scope. Identify the current GHG emissions rate of each audited emission unit at the time of the audit.

I.C.1.a.(i)(A) Control technologies and strategies must include, but are not limited to, fuel use, raw material use, energy efficiency improvements, preheating/heat reuse and strategic energy management options), and carbon capture and underground storage or utilization.

I.C.1.a.(ii) Eliminate technically infeasible control technologies and strategies.

I.C.1.a.(ii)(A) Notwithstanding Section I.C.1.a.(ii), the audit team must perform a feasibility assessment of carbon capture and underground storage or utilization technology for any single emissions unit evaluated with direct emissions of 100,000 tons per year or greater CO₂e in any of the previous five years as reported under Regulation Number 22, Part A and/or 40 CFR Part 98. The audit team shall include this analysis in the audit report.

I.C.1.a.(iii) Rank remaining emission unit control technologies and strategies in descending order based on the reduction in direct GHG emissions per ton of product or output.

- I.C.1.a.(iv) Perform a cost-effectiveness analysis on all control technologies and strategies for the emissions unit considering the full lifetime of the equipment. The cost-effectiveness analysis must include an estimate of the net levelized cost per ton of GHG emission reductions (\$/ton CO₂e) over the life of each proposed control method. The audit team must document in the audit report the discount rate, which is of no more than 8%, used for the cost-effectiveness analysis. The net levelized cost analysis should include, but is not necessarily limited to, the following costs and benefits:
 - I.C.1.a.(iv)(A) Engineering and design costs;
 - I.C.1.a.(iv)(B) Equipment costs, including installation;
 - I.C.1.a.(iv)(C) Available tax credits and/or incentive programs; and
 - I.C.1.a.(iv)(D) Changes in the annual costs resulting from the control technology/method including energy costs, operations and maintenance costs, and changes to productivity and/or product quality.
 - I.C.1.a.(v) Eliminate cost-prohibitive GHG reduction measures considered. GHG reduction measures with a cost-effectiveness of equal to or less than the social cost of GHGs cannot be eliminated as cost-prohibitive, except for a demonstrated, unreasonable burden on competitiveness as analyzed in Section I.C.1.a.(vi).
 - I.C.1.a.(vi) Consider the economic, energy, and environmental impacts arising from each option under consideration. In this case, economic reasonableness includes an analysis of the economic impact of the emission unit control option on the EITE stationary source's competitiveness within the marketplace. The audit team must document these determinations and associated analyses in the audit report.
 - I.C.1.a.(vii) Additional required documentation for all control technologies and strategies must include, but are not limited to, overall implementation cost, control efficiency, remaining useful life of the equipment, impacts to land use approvals, and a quantification of any co-benefits. The level of analysis conducted and documented shall consider the nature of the GHG BAECT measure and potential impacts of these additional criteria.
 - I.C.1.a.(viii) The GHG BAECT analysis may reference recently permitted GHG best available control technologies (BACT), operational or process limits in the EITE stationary source's air pollution permits or in the RACT/BACT/LAER Clearinghouse for similar operations as applicable.
- I.C.1.b. The energy BMP analysis.
- I.C.1.b.(i) Unless the EITE stationary source successfully demonstrates that it currently employs energy BMPs pursuant to Section I.C.1.b.(ii) or (iii) and provides the requisite supporting information pursuant to Sections I.C.1.b.(ii)(A) and (B) or I.C.1.b.(iii)(A) to I.C.1.b.(iii)(C), the audit team must analyze energy BMPs as follows:

- I.C.1.b.(i)(A) Identify all available energy efficiency measures for the specific energy consumption sources included in the audit scope. Any energy efficiency measure considered to be a GHG BAECT option but not recommended as GHG BAECT must be included in the energy BMP analysis for that GHG emission unit, as applicable. This analysis can exclude any control technologies that redefine the stationary source.
- I.C.1.b.(i)(B) Eliminate technically infeasible energy efficiency measures.
- I.C.1.b.(i)(C) Rank remaining energy efficiency measures based on the reduction in energy use per ton of final product manufactured at the facility.
- I.C.1.b.(i)(D) Perform a cost-effectiveness analysis on all energy efficiency measures considering the full lifetime of the measure. The cost-effectiveness analysis must include an estimate of the net levelized cost per energy consumption reduction over the life of the equipment. The audit team must document in the audit report the discount rate, which is of no more than 8%, used for the cost-effectiveness analysis. The net levelized cost analysis should include, but is not necessarily limited to, the following costs and benefits:
- I.C.1.b.(i)(D)(1) Engineering and design costs;
- I.C.1.b.(i)(D)(2) Equipment costs including installation;
- I.C.1.b.(i)(D)(3) Available tax credits and/or incentive programs; and
- I.C.1.b.(i)(D)(4) Changes in the following annual costs resulting from the control technology/method including energy costs, operations and maintenance costs, and changes to productivity and/or product quality.
- I.C.1.b.(i)(E) Eliminate cost-prohibitive energy efficiency measures considered. Energy efficiency measures with a cost-effectiveness equal to or under the social cost of GHGs cannot be eliminated as cost-prohibitive, except for a demonstrated, unreasonable burden on competitiveness shown in Section I.C.1.b.(i)(F).
- I.C.1.b.(i)(F) Consider the economic, energy, and environmental impacts arising from each measure remaining under consideration. In this case, economic reasonableness includes an analysis of the economic impact of the measure on the EITE stationary source's competitiveness within the marketplace. The audit team must document these determinations and associated analyses in the audit report.

- I.C.1.b.(i)(G) Additional required documentation for all analyzed measures include, but are not limited to, cost-effectiveness, remaining useful life of the equipment, impacts to land use approvals and any co-benefits. The level of analysis conducted and documented shall consider the nature of the energy efficiency measure and potential impacts of these additional criteria.
- I.C.1.b.(ii) In lieu of performing the energy BMP analysis, certification within 12 months of the audit date under the annual Federal Energy Star Program will be determined as employment of energy BMPs for the EITE stationary source. Annual Energy Star certification documentation for all years subsequent to the previous audit as well as the current certification must be included in the audit report and contain:
 - I.C.1.b.(ii)(A) Specific BMP energy efficiency measures the EITE stationary source used to achieve the Federal Energy Star Program certification; and
 - I.C.1.b.(ii)(B) The annual Energy Performance Indicator (EPI) benchmarking spreadsheet demonstrating a score of 75 or higher submitted with the Energy Star application.
- I.C.1.b.(iii) In lieu of performing the energy BMP analysis, registration to ISO 50001 within 12 months of the audit date will be determined as employment of energy BMPs for the EITE stationary source. Management system documentation must be included in the audit report and contain:
 - I.C.1.b.(iii)(A) Specific BMP energy efficiency measures the EITE stationary source used to achieve the ISO 50001 Program certification;
 - I.C.1.b.(iii)(B) Information on the energy management system including the Manual, Objectives and Goals, Energy Policy and results of the most recent energy management system audit; and
 - I.C.1.b.(iii)(C) The valid registration certificate.
- I.C.1.b.(iv) If an EITE stationary source fails to achieve the annual certification by the EPA Energy Star Program or registration to ISO 50001, the source must submit a compliance action plan to the Division within 90 days of the certification or registration expiration. The plan must include the EITE stationary source's plan and timeline to implement energy BMPs to either reacquire certification in the EPA Energy Star Program, reacquire ISO 50001 registration, or comply with the requirements in Section I.C.1.b.(i). The energy BMPs must be achieved within twelve months after the compliance action plan is approved by the Division.
- I.C.1.c. The GHG BAECT and energy BMP recommendation.
 - I.C.1.c.(i) The GHG BAECT recommendation will include:

I.C.1.c.(i)(A) Recommendations on the most effective direct GHG emissions control technology and strategy, or suite of technologies and strategies, for the GHG emissions unit analyzed as GHG BAECT;

I.C.1.c.(i)(B) A list of emissions control measures with a levelized cost less than or equal to \$0; and

I.C.1.c.(i)(C) Recommendations on GHG BAECT options that provide greater co-benefits to the surrounding communities where the top emission unit control technologies or strategies are comparable in terms of cost-effectiveness.

I.C.1.c.(i)(D) A calculation of the Non-GHG BAECT emissions. Non-GHG BAECT emissions are calculated by subtracting the reported emissions from units evaluated for GHG BAECT from the facility annual emissions at the time of the first audit. This shall be calculated as follows:

Non-GHG BAECT Emissions = Total direct emissions from the most recent year reported – (reported emissions from the units evaluated for GHG BAECT)

I.C.1.c.(ii) The energy BMP recommendation will include:

I.C.1.c.(ii)(A) Recommendations on the most effective energy efficiency measures for the energy consumption sources analyzed to be set as Energy BMPs.

I.C.1.c.(ii)(B) A list of energy efficiency measures with a levelized cost less than or equal to \$0;

I.C.1.c.(ii)(C) Recommendations on Energy BMP options that provide greater co-benefits to the surrounding communities where the top emissions unit control technologies or strategies are comparable in terms of cost effectiveness.

I.C.1.d. A plain-language summary of the audit findings, determinations, and recommendations in the top two languages spoken by the community surrounding the EITE stationary source. This summary shall include the list of GHG BAECT options for the emission units analyzed, how they were ranked and why they are being recommended.

I.C.1.e. Confidential business information must be clearly identified and be submitted in a separate, supplementary document to the audit report.

II. GHG BAECT and Energy BMP Determination

II.A. Within 60 days of receipt of the audit report, the Division will determine GHG BAECT and energy BMPs for the EITE stationary sources as follows:

II.A.1. The GHG BAECT determination may be based on the top-ranked control technology for the emission units in the audit report, or a suite of control technologies, at the Division's discretion, so long as:

- II.A.1.a. The cost of the suite of control measures, combined, would not be unduly burdensome to the facility; and
- II.A.1.b. The total annual CO₂ emissions reductions generated by the suite of measures is equal to or greater than the annual CO₂ emissions reductions generated by the top-ranked control technology according to the audit report.
- II.A.2. A GHG BAECT determination will be issued for each GHG emissions unit in the audit scope and include:
 - II.A.2.a. The Division's determination of specific control technologies and/or measures for that emission unit, including the current GHG emissions rate of each audited emission unit at the time of the audit;
 - II.A.2.b. All control technologies found under Section I.C.1.a. to have a net levelized cost less than or equal to \$0, where cost neutrality can be achieved no more than 5 years from operational date of the technology, except where the auditor has established that doing so would pose an unreasonable burden to the facility; and
 - II.A.2.c. The GHG emission rate per unit of final product at the EITE stationary source for the individual emissions units after implementation of the prescribed measures.
 - II.A.2.d. The final GHG BAECT and energy BMP intensity rate determination shall be issued for the EITE stationary source as a rate of total GHG emission (CO₂e) for the emissions units included in the audit scope per final product of the facility. This calculation shall be based on the determination(s) in Sections II.A.1. and II.A.3. (as applicable) and shall be calculated as follows:
 - II.A.2.d.(i) The Division may adjust the final GHG BAECT and energy BMP intensity determination based on a change to the GHG emission rate for any GHG BAECT measure after the GHG reduction measure is fully operational and tuned.
- GHG BAECT & Energy BMP Intensity Rate Determination = Σ (CO₂e per tons of facility product for each emission unit in audit scope).*
- II.A.3. The energy BMP determination will be issued for each energy consumption source in the audit scope and include:
 - II.A.3.a. The Division's determination of energy BMPs for the specific energy consuming equipment; and
 - II.A.3.b. Any energy efficiency measures found under Section I.C.1.b. to have a net levelized cost less than or equal to \$0 that are not included in the BAECT determination, unless the auditor has established that doing so would pose an unreasonable burden to the facility.
- II.B. The Division will hold one or more public meetings on the results of the final GHG BAECT and energy BMP determinations.
- II.C. Within 45 days of making its final GHG BAECT and energy BMP determinations, the Division will present the determinations at a regular meeting of the Commission. The Commission may approve the determinations or return them to the Division for further analysis. The Division will return to the Commission for final approval at its next regular meeting or as soon as practical.

III. Emission Reduction Requirements

III.A. All EITE stationary sources subject to this rule must reduce facility-wide GHG emissions by 5 percent.

III.A.1. EITE stationary sources annual emissions limitation shall be calculated as described in this Section III.A.1. Multiply the EITE stationary source's prescribed final GHG BAECT and energy BMP intensity rate determination, as determined by the Division under Section III. by the number of units produced in the calendar year, to result in a GHG BAECT mass emission number. If an EITE stationary source produces more than one product and, as a result, has more than one final GHG BAECT determination, the GHG BAECT mass emissions is calculated by multiplying the intensity rate for each product by the amount of product produced in the previous year, to calculate the GHG BAECT mass emissions for each product, and then adding the mass emissions for each product to calculate the GHG BAECT mass emissions. Add the Non-GHG BAECT emissions calculated under Section I.C.1.c.(i)(D) to the GHG BAECT emissions number, resulting in a mass-based emissions representing the facility-wide total GHG emissions (CO₂e), which serves as the baseline for the 5 percent mass-based reduction. Subtract the five (5) percent mass emissions reduction as required under Section III. This shall be calculated as follows:

*Annual Emission Limit for Compliance Year in tons CO₂e = (((GHG BAECT & Energy BMP Intensity Rate Determination) * (Compliance Year Facility Product)) + Non-GHG BAECT determination) * 0.95*

III.A.2. Beginning no later than the third year after each audit year, EITE stationary sources must demonstrate the additional mass-based five (5) percent GHG emission reduction and continued compliance with the reduction described in this section, by comparing the total mass GHG emissions limitation to the direct reported GHG emissions for the previous year. This is demonstrated through the annual compliance certification in Section V.A.

III.A.3. Where an EITE stationary source's approved compliance action plan requires that the plan be completed and operational after the third year after each audit year, the EITE stationary source must meet an interim annual mass emission limit achieving the required mass-based five (5) percent GHG emission reduction.

III.A.3.a. This interim annual five (5) percent mass reduction shall be calculated by multiplying the EITE stationary source's GHG BAECT and energy BMP intensity rate, as determined by the Division under Section II. by the number of units produced in the third calendar year, to result in a GHG BAECT mass emission number. Add the Non-GHG BAECT emissions to this number. Five (5) percent of this mass emissions total represents the mass GHG emissions reduction required in tons of CO₂e. The calculation is as follows:

*(((GHG BAECT & Energy BMP Intensity Rate Determination) * (Compliance Year Facility Product)) + Non-GHG BAECT determination) * 5%*

III.A.3.b. The interim annual emissions limitation is calculated by multiplying the current GHG emissions intensity rate as documented in the most recent audit by the compliance year facility product to result in a mass GHG emissions total. Add the Non-GHG BAECT emissions to this total, and then subtract the 5% reduction calculated in Section III.A.3.a. The calculation is as follows:

*((Current GHG Emissions Intensity Rate as documented in the audit report * Compliance Year Facility Product) + Non-GHG BAECT determination) – (((GHG BAECT & Energy BMP Intensity Rate Determination) * (Compliance Year Facility Product)) + Non-GHG BAECT Determination) * 5%)*

- III.B. If, at any point after the 2022 audit cycle and before 2030 an EITE stationary source achieves mass based GHG emission reductions equal to or greater than 20% below the source's 2015 GHG emissions baseline, the 5% emission reduction required under Section III.A. is considered satisfied through the year 2030, provided the 20% mass-based reductions are sustained as demonstrated through annual compliance certifications under Section V.A.1. An EITE stationary source meeting this requirement must continue to conduct the annual audits and otherwise comply with the requirements of this rule.

IV. Emission Reduction Requirement Compliance

- IV.A. The EITE stationary source must submit a compliance action plan within 120 days of the Commission's approval of the GHG BAECT, energy BMPs, and GHG BAECT and energy BMP intensity rate determination that includes the EITE stationary source's plan and timeline to comply with the annual mass emission limit, and interim mass emission limit in Section IV.A.2.d. as applicable and the energy BMPs determination.

- IV.A.1. The EITE stationary source may use the following emission reduction mechanisms to comply with the annual and interim emission limits:

IV.A.1.a. Actual, direct emissions reductions onsite at the EITE stationary source;
or

IV.A.1.b. Actual direct GHG emission reductions submitted in the form of GHG credits awarded by the Division. In order to produce GHG credits, the emissions reductions must be real, additional, verifiable, permanent and enforceable. GHG credits will only be issued in the tracking system after emission reduction have been demonstrated; or

IV.A.1.c. Installation and utilization at the facility of a retail distributed generation or net meter renewable energy project that reduces the GHG emissions from the EITE stationary source's electrical energy use, and for which the EITE stationary source has retired Renewable Energy Credits in Colorado generated that year, which demonstrate emissions reductions to comply with Section III.A. Utilization of Renewable Energy Credits for compliance cannot exceed the annual generation of the distributed generation system and it can only be used to satisfy the five (5) percent GHG emission reduction associated with Section III.A. and shall not be used for any other component of the annual emission limitation.

- IV.A.2. If the measure(s) determined to be GHG BAECT and/or energy BMPs for an emission unit also are anticipated to result in significant co-benefits, then the EITE stationary source must demonstrate in its compliance action plan that the co-benefits will be achieved through comparable reductions of the relevant harmful air pollutant(s) at the EITE stationary source. If an EITE stationary source demonstrates there are no material co-benefits from the implementation of the measure(s) determined to be GHG BAECT and/or energy BMPs, these requirements do not apply.

- IV.B. Beginning no later than the third year after each audit year, EITE stationary sources must demonstrate the additional mass-based five (5) percent GHG emission reduction described in this Section V. through the annual compliance certification in Section V.A.1.

- IV.C. When considering compliance options with similar or the same cost-effectiveness, the EITE entity must give increased priority to GHG reduction initiatives that would produce co-benefits to the neighboring communities surrounding the EITE stationary source.
- IV.D. The Division will review the compliance action plan for approval.
- IV.E. EITE stationary sources must comply with the compliance action plan once approved by the Division.
- IV.F. If in any calendar year an EITE stationary source achieves reported emissions lower than the annual emissions limitation, the Division will award the EITE stationary source GHG credits equal to the difference between the annual emissions limitation and reported emissions. GHG credits will only be issued after emission reductions have been demonstrated.
- IV.G. If in any calendar year an EITE stationary source fails to achieve the annual emissions limitation, the owners and operators may remedy this noncompliance by surrendering in the EITE entity's compliance account sufficient GHG credits so as to reduce the actual emissions to the EITE annual emissions limitation.

V. Reporting

- V.A. Owners or operators of EITE stationary sources must submit an annual report to the Division by May 1 of each year (beginning May 1, 2026) that includes:
 - V.A.1. Annual Compliance Certification Requirements. The account representative of each EITE stationary source must submit a compliance certification to the Division or its agent that certifies:
 - V.A.1.a. The EITE stationary source's annual GHG emissions, in metric tons, for the previous calendar year;
 - V.A.1.b. The total production at the EITE stationary source, in units that are consistent with the GHG BAECT and energy BMP intensity rate determination, for the previous year;
 - V.A.1.c. The EITE stationary source's GHG emissions rate per product for the previous calendar year and the GHG BAECT and energy BMP intensity rate determination.
 - V.A.1.d. The EITE stationary source's annual emissions limitation for the previous year;
 - V.A.1.e. The difference, if any, between the EITE stationary source's actual total direct GHG emissions for the previous year and the EITE stationary source's annual emissions limitation for the previous year.
 - V.A.1.f. For EITE stationary sources that are complying through GHG credits pursuant to Section VI.A.1.b.:
 - V.A.1.f.(i) In the event that the EITE stationary source has actual total emissions that exceed its annual emissions limitation, the compliance certification shall specify the GHG credits that are to be surrendered from the EITE stationary source's compliance account sufficient to meet the EITE stationary source's annual emissions limitation.

- V.A.2. Current project status to implement GHG BAECT or energy BMPs contained in a compliance action plan under Section IV.A.;
- V.A.2.a. The current status of measures to achieve comparable co-benefits required under Section IV.A.2.
- V.A.3. If the EITE stationary source is determined to be employing energy BMPs through certification to the Federal Energy Star Program or the ISO 50001 standard, the EITE stationary source must submit the information in accordance with Section I.C.1.b.;
- V.A.4. Instances of noncompliance with the Division's approved GHG BAECT and energy BMPs determination, compliance action plan, reason(s) for noncompliance, and actions taken or planned to return to compliance; and
- V.A.5. All information necessary for the Division to confirm the EITE stationary source's emission rate in the prior year.
- V.B. In addition to the annual report, owners and operators of EITE stationary sources must submit a final audit update to the Division within 60 days of the operation of GHG BAECT and/or energy BMPs, which includes verification that all GHG BAECT and energy BMP measures established in Section II. are operational.

VI. Recordkeeping

- VI.A. EITE stationary sources must maintain records for a period of ten (10) years and make records available to the Division upon request, including:
 - VI.A.1. Division approved audit plan.
 - VI.A.2. Final audit reports.
 - VI.A.3. Commission approved GHG BAECT and energy BMP determinations.
 - VI.A.4. Annual Compliance Certificate.
 - VI.A.5. Approved compliance action plans and records reasonably necessary to demonstrate compliance with the approved plan.
 - VI.A.6. Current certification or registration documentation to applicable standards to show continued compliance with Section I.C.1.b.(ii) or (iii).
 - VI.A.7. If the final GHG BAECT is determined to have an operation date that is beyond the next 5-year audit, the EITE stationary source must maintain records of project management documentation related to the implementation of the GHG BAECT measure including project status, timeline, expected operational date as proposed and approved in the compliance action plan.

PART D Greenhouse Gas Credit Trading

I. Establishment and Maintenance of Accounting and GHG Crediting and Tracking System and Accounts

- I.A. By December 1, 2024, the Division or its agent will establish a GHG crediting and tracking system.

- I.B. Upon receipt of the registration requirements under Part D, Section II., the Division or its agent will establish one (1) credit account for each manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator in the GHG crediting and tracking system.
- I.C. The Division will assign a specific identifier in the GHG crediting and tracking system to a manufacturing stationary source located within a disproportionately impacted community.
- I.D. If there is no activity on a credit account(s) in the GHG crediting and tracking system for eighteen (18) months or longer, the Division may purge the unused credit account(s) from the GHG crediting and tracking system as long as the Division provides written notice to the account representative(s) of the intended purge and either the account representative(s) does not respond to the Division's written notice of the intended purge in writing within a reasonable time or the account representative(s) agree to the purge in writing.

II. Registration for the Greenhouse Gas Crediting and Tracking System

- II.A. All manufacturing stationary sources; midstream segment companies; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator must submit an application for an environmental identifier (environmental ID) with the Division if the entities does not already has an environmental ID.
- II.B. All manufacturing stationary sources; midstream segment companies; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator must submit a registration application for a credit account to the Division or its agent as follows.
 - II.B.1. All GEMM 2 facilities must submit a registration application for a credit account in the GHG crediting and tracking system to the Division or its agent by December 31, 2024, or within thirty (30) days after the GEMM 2 facility first reports annual direct GHG emissions equal to or greater than the 25,000 metric tons of CO₂e threshold, whichever occurs later.
 - II.B.2. All EITE stationary sources must submit a registration application for a credit account in the GHG crediting and tracking system to the Division or its agent within thirty (30) days after the Commission approves the EITE stationary sources' GHG BAECT and energy BMP determination, or within thirty (30) days after the EITE stationary source first reports annual direct GHG emissions equal to or greater than the 25,000 metric tons of CO₂e threshold, whichever occurs later.
 - II.B.3. All midstream segment companies must submit a registration application for a credit account in the GHG crediting and tracking system to the Division or its agent by December 1, 2028, or within ninety (90) days of becoming a midstream segment company subject to Regulation Number 7, Part B, Section VII., whichever occurs later.
 - II.B.4. All gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator must submit a registration application for a credit account in the GHG crediting and tracking system to the Division or its agent before they may submit an application for recovered methane credits or participate in a credit transaction in the GHG crediting and tracking system.
- II.C. In addition and in connection with each registration application, registration applicants must include other information that the Division deems necessary, including but not limited to the following information.

- II.C.1. Legal name(s), physical and mailing addresses, contact information, date and place of incorporation, and any identification number assigned by the incorporating agency of the owner or operator that controls the manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator;
- II.C.2. Legal name(s), mailing addresses, and contact information of the directors and officers with authority to make legally binding decisions on behalf of the manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or owner or operator of a recovered methane project, including any partners with over ten (10) percent control of a partnership that owns or controls the manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator, including any individual or entity doing business as the limited partner or the general partner;
- II.C.3. Legal name(s) and contact information of person(s) with over ten (10) percent control of the voting rights attached to all the outstanding voting securities of the entity that owns or controls the manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator;
- II.C.4. A business number, if one has been assigned to the entity by a Colorado state agency, to the entity that owns or controls the manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator;
- II.C.5. NAICS code, AIRS ID, and EPA's Greenhouse Gas Reporting Program Facility ID for the manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator;
- II.C.6. A government-issued taxpayer or Employer Identification Number or, for entities located in the United States, a U.S. Federal Tax Employer Identification Number, if assigned, for the entity that owns or controls the manufacturing stationary source; midstream segment company; and gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator;
- II.C.7. A confirmation of whether the manufacturing stationary source or gas distribution utility, municipal gas distribution utility, small gas distribution utility, or recovered methane project owner or operator is located within a disproportionately impacted community using Colorado EnviroScreen; and
- II.C.8. The midstream segment company's registration application must include an attachment confirming whether and which of the company's facilities are located within a disproportionately impacted community, as identified using Colorado EnviroScreen Version 1.
- II.D. Any individual who requires access to the GHG crediting and tracking system on behalf of a manufacturing stationary source or midstream segment company or midstream segment company must register as a user in the GHG crediting and tracking system.
- II.D.1 Each manufacturing stationary source and midstream segment company must designate one primary account representative and at least one, and up to four, alternate account representatives.

- II.D.2. Registering users must provide their legal name(s) and contact information when applying to become an account representative.
- II.D.3. Each account representative shall represent, and by his or her representations, actions, inactions, or submissions, legally bind the manufacturing stationary source or midstream segment company in all matters relating to Regulation Number 27. Any representation, action, inaction, or submission by any authorized account representative will be deemed to be a representation, action, inaction, or submission by the source.

III. Greenhouse Gas Credit Trading Requirements

III.A. GHG credits will be generated as follows.

III.A.1. For the purpose of trading GHG credits between EITE facilities, EITE facilities will generate one (1) GHG credit per metric ton of CO₂e for each metric ton of CO₂e that the relevant facility's annual direct GHG emissions are less than its annual emissions limitation in the relevant year.

III.A.2. For the purpose of trading GHG credits between GEMM 2 facilities:

III.A.2.a. Subject to Section III.A.2.a.(i), in any compliance year, a GEMM 2 facility will generate one (1) GHG credit for each metric ton of CO₂e that the facility's annual direct GHG emissions are less than its GEMM 2 annual GHG emissions requirement for 2030.

III.A.2.a.(i) In any compliance year, a GEMM 2 facility with a production-related adjusted baseline will generate one (1) GHG credit for each metric ton of CO₂e that the facility's annual direct GHG emissions are less than the result of the below calculation:

*(Higher of the GEMM 2 facility's 2021 or 2022 emissions, as reported under Regulation Number 22) * (1 - 2030 GEMM 2 facility GHG percentage reduction requirement as specified in Part B, Section I.A.)*

III.A.2.b. Beginning in the 2031 compliance year, a GEMM 2 facility may generate one (1) GHG credit per one (1) metric ton of CO₂e quantifiably reduced through offsite direct air carbon capture projects, provided, however, that such projects may be used for compliance with this Regulation Number 27 only after the Division approves a protocol governing the implementation of such projects.

III.A.3. GHG credit trading between EITE and GEMM 2 facilities will be allowed beginning in 2025 subject to Division guidance to be published no later than December 1, 2024.

III.B. The Division will issue GHG credits to manufacturing stationary sources as follows.

III.B.1. On an annual basis beginning with the 2024 compliance year, and by the first Tuesday of May 2025 and on the first Tuesday of each May thereafter, the Division will credit the GHG crediting and tracking system credit account of an operating GEMM 2 facility that has submitted all reports in accordance with Part B, Section IV.A, as verified by the Division, with GHG credits equal to the difference between the facility's annual direct GHG emissions in the applicable compliance year and its GEMM 2 annual GHG emissions requirement for 2030 as established in Part B, Section I. A GEMM 2 facility which has ceased operations may not generate GHG credits.

- III.B.2. The Division will annually issue GHG credits in the GHG crediting and tracking system to an EITE stationary source to the extent its annual direct GHG emissions in the applicable compliance year are less than its annual emissions limitation in the applicable compliance year, as a result of permanent, real, and verifiable emission reduction measures implemented at the EITE stationary source.
- III.B.3. Each GHG credit issued by the Division in the GHG crediting and tracking system will be uniquely identifiable.
- III.B.4. By the third Tuesday of May 2025 and each third Tuesday of May thereafter, the Division will publish on its website the total amount of any GHG credits credited collectively to the compliance accounts of manufacturing stationary sources for the previous calendar year, and the cumulative surplus or shortfall of credits for that vintage year.
- III.C. A GHG credit will expire three (3) years after its issuance date in the GHG crediting and tracking system unless it is retired prior to expiration in the GHG crediting and tracking system. An expired GHG credit may no longer be retired, sold, or transferred in the GHG crediting and tracking system.
- III.D. To avoid double-counting of GHG emission reductions, GHG credits generated under this Regulation Number 27, Part D, may not be sold into or used in any carbon or GHG offset registry or trading market outside of the GHG crediting and tracking system.
- III.E. Subject to Part D, Sections III.A.3. and Regulation Number 7, Part B, Section VII.F.6.f., any manufacturing stationary source or midstream segment company may sell GHG credits to any other manufacturing stationary source or midstream segment company upon such terms as agreed upon between the manufacturing stationary source and midstream segment company at any time, except to the extent the GHG credits are offered for sale pursuant to an auction under Part D, Section IV. Within thirty (30) days of completing any GHG credit sale, the manufacturing stationary sources or midstream segment company participating in the transaction will report to the Division, through a Division-approved process, the quantity and price or other consideration, for each vintage year(s) of GHG credits sold, and the Division shall post on its website the parties, quantity, and vintage year(s) of GHG credits sold within thirty (30) days of receiving such report. The Division may elect to publish pricing information from such reports.
 - III.E.1. Upon reporting of a GHG credit transaction, the GHG credits subject to the trade will be transferred from the selling party's compliance account to the buying party's compliance account within the GHG crediting and tracking system.
 - III.E.2. In selling GHG credits, the seller warrants and represents that the credits generated are based on an accurate reporting of GHG emissions for the relevant year. Buyers of GHG credits buy GHG credits with this assurance.

IV. Annual GHG Credit Auctioning

- IV.A. On June 30, 2025 and each June 30 or first business day thereafter, the auction administrator will administer a voluntary GHG credit auction. Only manufacturing stationary sources may participate in any GHG credit auction to buy or sell GHG credits. Beginning in 2028 with GHG credit vintage year 2027, midstream segment companies may participate in the GHG credit auction to buy or sell GHG credits.
 - IV.A.1. The auction administrator shall create a publicly available website for publishing information related to the annual auction. The website shall contain the following information:

- IV.A.1.a. General guidance on the process, including how bids and offers are used to determine the auction settlement price and the total quantity of GHG credits sold at auction.
- IV.B. Beginning the third Tuesday of May 2025 and each third Tuesday of May thereafter, the auction administrator shall publish a notice of auction on its website, with the following information:
 - IV.B.1. Auction application requirements and instructions;
 - IV.B.2. The form and manner for submitting bids;
 - IV.B.3. The procedures for conducting the auction; and
 - IV.B.4. The time, date, and procedures for a second round of the auction, if necessary, pursuant to Section IV.I.
- IV.C. By May 31, 2025 and every May 31 thereafter, any manufacturing stationary source that desires to participate in the auction, for each vintage year of GHG credits, as a bidder or offeror must inform the auction administrator in writing in order to participate and submit bids or offers, respectively, in the auction. In order to participate and submit bids or offers in the auction, midstream segment companies must inform the auction administrator in writing by May 31, 2029, and every May 31 thereafter. Failure to provide such notice for a vintage year of GHG credits precludes the manufacturing stationary source or midstream segment company from participating in that year's annual auction process for the respective vintage year.
 - IV.C.1. A manufacturing stationary source or midstream segment company that offers to sell GHG credits in one or more vintage years in any auction is not eligible to bid to purchase GHG credits from the same vintage years in the same auction, but is eligible to bid on GHG credits from vintage years different than those offered for sale.
 - IV.C.2. A manufacturing stationary source or midstream segment company that bids to buy GHG credits in one or more vintage years in any auction is not eligible to offer to sell GHG credits from the same vintage years in the same auction, but is eligible to offer GHG credits for sale from vintage years different than those bid for purchase.
- IV.D. Three (3) business days after receiving notices of intent to participate, the auction administrator will determine whether a sufficient number of bidders and offerors have given notice of their intent to participate in the annual auction. A sufficient number requires more than zero offerors and more than zero bidders. If there is a sufficient number of bidders and offerors, then the auction administrator will publish a notice stating that the auction will take place that year and will provide the number of bidders and offerors that have provided a notice of intent to participate. If there is not a sufficient number of bidders and offerors, then the auction administrator will publish a notice on its website canceling the auction for that year, and the timeline in the remaining sections of this Part D shall not apply.
- IV.E. Auction window. Five (5) business days after May 31 of the relevant year, the auction administrator will accept bids and offers from bidders and offerors, respectively, for each vintage year of GHG credits through the GHG crediting and tracking system managed by the auction administrator, and the auction administrator shall allow bids and offers to be submitted by no later than June 15 of the relevant year.
 - IV.E.1. The auction bidding and offering window may be delayed, rescheduled, or canceled due to technical systems failures.

- IV.E.1.a. If technical systems failures cannot be resolved and a bidding window cannot be rescheduled to meet the requirements of this section, then the auction administrator will cancel the auction bidding window.
- IV.F. All bids for GHG credits will be considered a binding commitment for the purchase of GHG credits under the rules of the auction. All offers for GHG credits submitted for sale in the auction will be considered a binding commitment for the sale of GHG credits under the rules of the auction, and by offering GHG credits for sale, potential sellers warrant and represent that the credits generated are based on an accurate reporting of GHG emissions for the relevant year.
- IV.G. Auction Format
 - IV.G.1. A separate auction will be held for each vintage year of GHG credits, and each auction will consist of a single round of bids and offers for GHG credits of the applicable vintage year.
 - IV.G.2. Each individual manufacturing stationary source or midstream segment company may only act as a bidder or offeror of GHG credits for an individual vintage year in the annual auction, consistent with Section IV.C.
 - IV.G.3. Bidders submit bids and offerors submit offers.
 - IV.G.4. Each bidder may submit multiple bids for each vintage year of GHG credits.
 - IV.G.5. Each offeror may submit multiple offers for each vintage year of GHG credits.
 - IV.G.6. Bids and offers will be sealed and given confidential treatment.
 - IV.G.7. Bids and offers must specify the quantity of GHG credits to be transacted and must be submitted as multiples of tens (10) or hundreds (100) GHG credits.
 - IV.G.8. Bids and offers must be submitted in U.S. dollars and whole cents, on a per-GHG credit basis.
 - IV.G.9. Bids and offers must specify the vintage year of the GHG credits for which they are made.
- IV.H. Determination of Winning Bidders and Settlement Price. For each auction of each vintage year of GHG credits, the following process shall be used to determine a single auction settlement price, allocations of GHG credits sold to bidders, and proceeds from such sales to offerors:
 - IV.H.1. Each bid will consist of a price and the quantity of GHG credits of the applicable vintage year, in multiples of tens (10) or hundreds (100) GHG credits, desired at that price.
 - IV.H.2. Each offer will consist of a price and the quantity of GHG credits of the applicable vintage year, in multiples of tens (10) or hundreds (100) GHG credits, available for sale at that price.
 - IV.H.3. Each bidder may submit multiple bids for each vintage year of GHG credits.
 - IV.H.4. Each offeror may submit multiple offers for each vintage year of GHG credits.
 - IV.H.5. Beginning with the highest bid price, bids from each bidder will be considered in declining order by price.

- IV.H.6. Beginning with the lowest offer price, offers from each offeror will be considered in ascending order by price.
- IV.H.7. The auction administrator will develop a Division-approved protocol for determining which GHG credits are exchanged and at what price through the auction, which will be every GHG credit for which a bid price exceeds or is equal to an offer price associated with the GHG credit considered in the order prescribed in Sections IV.H.5 and IV.H.6 offered into the auction.
 - IV.H.7.a. The protocol, through the single auction settlement price, must ensure that no bidder pays more per GHG credit than the bidder bid for a particular GHG credit, and no offeror receives less per GHG credit than the offeror offered for a particular GHG credit.
 - IV.H.7.b. The protocol will also provide a method for resolving tie bids and offers in a reasonable and equitable manner. The protocol shall also address, in a reasonable and equitable manner, circumstances where the lowest bid price exceeds the offer price at that quantity of total credit sales.
 - IV.H.7.c. Under the protocol, the auction administrator will determine the auction settlement price, and it will determine from which winning offeror(s) a winning buyer will purchase GHG credits.
- IV.I. Additional auction. If, in the first round of the auction for any vintage year's auction pursuant to Sections IV.G and IV.H, the total GHG credits sold through the auction is less than 50% of the total GHG credits contained cumulatively in all the offers submitted to the auction for the applicable vintage year, then there will be an additional auction round for any such vintage year of remaining GHG credits not sold in the first auction round.
 - IV.I.1. Within five (5) business days after the first auction, the auction administrator shall publish the auction settlement price from the first round of the auction for any vintage year, the highest and lowest bid price, and the highest and lowest offer price, the total number of GHG credits offered, and the median of all bid prices and offer prices, from the initial auction round.
 - IV.I.2. There shall only be one additional auction round. Bids and offers for the additional auction shall be received by the auction administrator no later than three (3) business days after the publication of the results of the first auction pursuant to Section IV.I.1, and the additional auction shall be held within five (5) business days after publication of the results of the first auction pursuant to Section IV.I.1.
- IV.J. Actions following auction.
 - IV.J.1. Within two (2) business days following an auction or any additional auction round held pursuant to Section IV.I, the auction administrator will notify, by vintage year:
 - IV.J.1.a. Each winning bidder of the auction settlement price, the number of GHG credits that the bidder purchased, the bidder's total purchase cost, and to which sellers of GHG credits to direct payment.
 - IV.J.1.b. Each winning offeror of the auction settlement price, the number of GHG credits that the offeror sold, the offeror's total sales revenue, and from which bidders to expect payment.

- IV.J.2. Within fifteen (15) business days following an auction or any additional auction round held pursuant to Section IV.I, the auction administrator will publish the following information:
 - IV.J.2.a. The identity of the manufacturing stationary sources and midstream segment companies that were bidders, but shall not publish the bids made by such entity nor the total quantity of GHG credits purchased by such entity;
 - IV.J.2.b. The identity of the manufacturing stationary sources and midstream segment companies that were offerors, but shall not publish the offers made by such entity nor the total quantity of GHG credits sold by such entity;
 - IV.J.2.c. The Auction settlement price; and
 - IV.J.2.d. Aggregated or distributional information on bids, offers, and GHG credit purchases.
- IV.J.3. To transfer the GHG credits sold at auction from the respective seller's credit account to the respective buyer's credit account, the seller and buyer shall provide notice to the Division within sixty (60) days of receiving the notification from the Division that they were a winning offeror or bidder requesting the GHG credits be transferred.

PART E Statements of Basis, Specific Statutory Authority and Purpose

I. Adopted: October 22, 2021

(Removed from Regulation Number 22 and placed in Regulation Number 27 April 20, 2023)

Revisions to Regulation Number 22, Part B, Section II.

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-103(4), C.R.S., the Colorado Air Pollution Prevention and Control Act, §§ 25-7-110 and -110.5, C.R.S., and the Air Quality Control Commission's ("Commission") Procedural Rules, 5 C.C.R. §1001-1.

Basis

In HB 19-1261, now codified in part at §§ 25-7-102(2) and -105(1)(e), C.R.S., the General Assembly declared that "[c]limate change adversely affects Colorado's economy, air quality and public health, ecosystems, natural resources, and quality of life[.]" acknowledged that "Colorado is already experiencing harmful climate impacts[.]" and that "[m]any of these impacts disproportionately affect" certain disadvantaged communities. § 25-7-102(2), C.R.S. The General Assembly also recognized that "[b]y reducing greenhouse gas pollution, Colorado will also reduce other harmful air pollutants, which will, in turn, improve public health, reduce health care costs, improve air quality, and help sustain the environment." § 25-7-102(2)(d), C.R.S.

Consequently, the General Assembly updated Colorado's statewide greenhouse gas (GHG) pollution reduction goals so as to achieve a 26% reduction of statewide GHG by 2025; 50% reduction by 2030; and 90% reduction by 2050 as compared to 2005 levels. § 25-7-102(2)(g), C.R.S. Statewide GHG pollution is defined as "the total net statewide anthropogenic emissions of carbon dioxide [(CO₂)], methane [(CH₄)], nitrous oxide [(N₂O)], hydrofluorocarbons [(HFCs)], perfluorocarbons [(PFCs)], nitrogen trifluoride [(NF₃)], and sulfur hexafluoride [(SF₆)] expressed as carbon dioxide equivalent [(CO₂e)] calculated using a methodology and data on radiative forcing and atmospheric persistence deemed appropriate by the commission." § 25-7-103(22.5), C.R.S.

§ 25-7-105(1)(e), C.R.S., sets forth the framework for developing GHG abatement rules consistent with the statewide GHG pollution reduction goals in § 25-7-102(2)(g), C.R.S. This provision grants the Commission broad authority to regulate GHG emissions in order to accomplish these goals.

In order to evaluate the utilization of, and potential emissions reductions from, GHG best available emission control technologies (BAECT) and best available energy efficiency practices (referred to as best management practices or Energy BMP) in energy-intensive trade-exposed (EITE) stationary sources, the Commission adopted in Regulation Number 22, Part B, Section II, rules governing emission control and energy audits from these sources and requiring a five percent reduction in GHG emissions therefrom.

Specific Statutory Authority

The Colorado Air Pollution Prevention and Control Act (Act), specifically § 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102, C.R.S., and that are necessary for the proper implementation and administration of the Act.

§ 25-7-105(1)(e), C.R.S., authorizes the Commission to promulgate implementing rules and regulations consistent with the statewide GHG pollution reduction goals in § 25-7-102(2)(g), C.R.S. In adopting GHG abatement strategies and implementing rules, the Commission is authorized to take into account other relevant laws and rules to enhance efficiency and cost-effectiveness and solicit input from other state agencies and stakeholders on the advantages of different statewide GHG pollution mitigation measures. § 25-7-105(1)(e)(II) and (IV), C.R.S.

Implementing rules may include regulatory strategies that incentivize development of renewable resources and “enhance cost-effectiveness, compliance flexibility, and transparency around compliance costs.” § 25-7-105(1)(e)(V), C.R.S. Further, in promulgating such implementing rules, the Commission is to consider many factors, including, but not limited to: health, environmental, and air quality benefits and costs; the relative contribution of each source or source category to statewide GHG pollution; equitable distribution of the benefits of compliance; issues related to the beneficial use of electricity to reduce GHG emissions; and whether greater or more cost-effective emission reductions are available through program design. § 25-7-105(1)(e)(VI), C.R.S.

§ 25-7-105(1)(e)(IX), C.R.S., authorizes the Commission to require energy-intensive, trade-exposed [(EITE)] stationary sources, “to execute an energy and emission control audit, according to criteria established by the [C]ommission, of the source’s operations every five years through at least 2035.” The intent of the audit is to determine whether covered sources are employing “best available emission control technologies for [GHG] emissions [(GHG BAECT)] and best available energy efficiency practices [(Energy BMP)]”.

§ 25-7-105(1)(e)(XIII), C.R.S., adopted in House Bill 21-1266, directs the Commission to “require a five percent reduction in the [GHG] emissions associated with [EITE] stationary sources that currently employ [GHG BAECT and Energy BMPs], as determined by the Commission, pursuant to [§ 25-7-105(1)(e)(IX), C.R.S.]”.

§ 25-7-106, C.R.S., provides the Commission “maximum flexibility in developing an effective air quality program and [promulgating] such [a] combination of regulations as may be necessary or desirable to carry out that program.” § 25-7-109(1), C.R.S., authorizes the Commission to adopt and promulgate emission control regulations that require the use of effective practical air pollution controls for each type of facility, process, or activity which produces or might produce significant emissions of air pollutants. An “emission control regulation” may include “any regulation which by its terms is applicable to a specified type of facility, process, or activity for the purpose of controlling the extent, degree, or nature of pollution emitted from such type of facility, process, or activity. . . .” § 25-7-103(11), C.R.S. Emission control regulations may pertain to any chemical compound including GHG pollution. See § 25-7-109(2)(c), C.R.S.

Purpose

The Commission adopted Regulation Number 22, Part B, Section II to give effect to the requirements of §§ 25-7-105(1)(e)(IX) and (XII)(B), C.R.S., and to further the reduction of statewide GHG pollution consistent with § 25-7-102(2)(g), C.R.S., as applicable to EITE stationary sources.

§ 25-7-105(1)(e)(IX), C.R.S., authorizes the Commission to require EITE stationary sources, “to execute an energy and emission control audit, according to criteria established by the [C]ommission, of the source’s operations every five years through at least 2035.” The intent of the audit is to determine whether covered sources are employing GHG BAECT and Energy BMPs. The last audit shall occur in the year 2037.

The audit and GHG emission reduction requirements contained in Part B, Section II are applicable only to stationary sources that principally engage in defined manufacturing activities and have direct GHG emissions equal to or greater than 50,000 tons CO₂e per year as reported under 40 CFR, Part 98 and/or Part A of this Regulation Number 22. Based on 2019 data, there are only four stationary sources meeting these requirements: EVRAZ Rocky Mountain Steel's mill in Pueblo, with reported GHG emissions of 305,674 tons CO₂e; CEMEX Construction Materials South's Lyons Cement Plant, with reported GHG emissions of 268,643 tons CO₂e; GCC Rio Grande's cement manufacturing plant in Pueblo, with reported GHG emissions of 743,403 tons CO₂e; and Holcim-Lafarge's cement plant in Florence, with reported GHG emissions of 985,222 tons CO₂e.

The fifth highest GHG emission sources in one of these enumerated manufacturing activities reported GHG emissions of approximately 27,000 tons CO₂e in 2019. The four EITE stationary sources over 50,000 tons CO₂e per year in 2019 remained the only sources over this threshold in 2020 as well. It is notable that, based on 2019 reported data, these four stationary sources contribute approximately 54% of Colorado industrial and manufacturing sector's GHG emissions. Accordingly, this emissions threshold guarantees that the state's largest EITE sources of GHG emissions are employing GHG BAECT and Energy BMPs in their manufacturing processes while appropriately limiting the regulatory burden on smaller sources.

Critical to the success of the GHG BAECT and Energy BMP audit program is the selection of sufficiently rigorous audit protocols and qualified auditors. Accordingly, Section II.C.2. and the associated definitions in Section II.B. establish the minimum qualifications for the auditor and audit team and the criteria and processes by which the audits are to be performed. Under Section II.C.2.a.(v)(A), the Division can reject an auditor if the auditor has a previous or existing relationship with the source that is so pervasive that the auditor would be unable to conduct the audit in an unbiased and independent manner. The Division should consider the extent to which the auditor has advocated on behalf of the EITE stationary source before the Commission; whether the relationship is currently or recently pervasive, compared to a relationship primarily in the past; whether the auditor has primarily acted as an advocate for the EITE stationary source as opposed to a scientific or technical advisor; and whether the work conducted on behalf of the EITE stationary source would, in any way, inhibit the auditor's ability to conduct an independent and unbiased audit.

Prior to executing an audit, the EITE stationary source must submit to the Division for its review and approval of the audit plan, which identifies the audit scope, the audit team and any standards or protocols planned for use in the audit. The Division's review should ensure that the plan is sufficiently rigorous and meets or exceeds national and/or international standards for such audits. Examples of sufficiently rigorous accounting and audit protocols include, but are not necessarily limited to, the GHG Protocol's Corporate Accounting and Reporting Standard (more information available at <https://ghgprotocol.org/corporate-standard>) and the International Organization for Standardization's (ISO) 14064 and 50001 series (more information available at <https://www.iso.org/home.htm>).

As set forth in Section II.C.2., the energy and emissions control audit will analyze GHG BAECT for the EITE stationary source's emissions units that emit the top 80% of the stationary source's GHG emissions, and any individual emissions source that represents more than 2% of the emissions from the EITE stationary source. This audit scope is determined by the Division to be broad enough to capture all GHG emission sources at the EITE stationary source except for those considered "de minimis" and provide for a thorough audit of the emissions at the facility.

The Energy BMP audit will assess all emission units that account for 80% of the source's energy consumption. This scope ensures that the audit captures the largest emitting and energy consuming emissions units and that the majority of the EITE stationary source's emissions are examined. It is expected that there may be overlap between these two scopes, which is addressed in Section II.C.3.a.(ii)(A)(1).

Section II.C.3. establishes the audit reporting requirements and detailed steps for the GHG BAECT and Energy BMP analyses. These analyses are conducted on a case-by-case basis for the EITE stationary source under review and incorporate certain objective criteria for the auditor and Division to consider. The GHG BAECT analysis and prioritization process will consider technical feasibility of the control for the stationary source, the estimated emission reductions realized with each measure, cost-effectiveness and environmental, economic, and energy impacts. Additionally, consideration should be given to other factors such as existing facility permit limits, limits and emissions data available for similar operations, and air pollution co-benefits to local communities.

Under Section II.C.3.a.(i)., the control technologies and strategies considered during the GHG BAECT analysis should consider fuel switching, waste to heat options, strategic energy management (SEM) options and carbon capture and underground storage or utilization (CCUS). In evaluating fuel-switching as a control technology, the source should consider switching from coal and petroleum coke to a lower-carbon fuel. In evaluating waste to heat options, the source should consider changing raw material inputs in production processes. In evaluating waste to heat options, the source should consider preheating and heat re-use. The Commission understands that certain of these technologies may not yet be ready for employment as GHG BAECT or Energy BMPs at the stationary sources and expects them to be evaluated for technical feasibility if they are “available.”

The cost-effectiveness evaluation will include the full lifetime of the measure under consideration and all “net levelized” costs to account for costs and costs avoided—or benefits—that may result from adopting a particular control or efficiency measure. These factors are described in further detail at Section II.C.3.a.(i)(D) for GHG BAECT and II.C.3.a.(ii)(A)(4) for Energy BMPs and should include changes in the annual costs resulting from the control technology/method including energy costs, operations and maintenance costs, productivity (e.g. increased production rate or reduced down-time), product quality (e.g. improved quality or reduced rate of rejects/bad batches). As detailed in Part B, Section II.C.3.a.(i)(E) for GHG BAECT and Section II.C.3.a.(ii)(A)(5) for Energy BMPs, measures identified in the audit as technically feasible cannot then be eliminated as cost-prohibitive if the cost-effectiveness of the measure is equal to or less than the avoided social cost of GHGs. Further, any direct or indirect energy, economic and environmental impacts for each potential GHG control measure is to be considered. Economic considerations may include a comparison to direct competitors and international markets for the EITE source’s final product (i.e. steel or cement). Environmental considerations should include any benefits or detriments that may result from a particular measure. Energy considerations should include direct and indirect energy efficiency benefits or detriments, including changes in demand for offsite electric generation.

The social cost of GHG’s cost-effectiveness comparison mechanism ensures that sufficient weight is afforded to the full spectrum of climate impacts from GHG emissions that could be controlled through technically feasible means. As set forth in Part B, Section II.B.41, the social cost of greenhouse gases to be used aligns with that established in § 25-7-110.5(4)(f), C.R.S., through the adoption of House Bill 21-1266. The social cost of GHGs to be used in each GHG BAECT and Energy BMP audit must be consistent with this definition and the Division will review the social cost of GHGs used in the audit recommendations to ensure the values are correctly calculated. Using the social cost of GHGs is an appropriate, objective measure for evaluating potential costs avoided when analyzing the “cost-prohibitiveness” of effective, technically feasible control measures and balancing that against other environmental, energy, or economic impacts, such as competitiveness.

This balancing serves to recognize the externalities resulting from GHG emissions that could be avoided through the use of potentially costly control measures and the limited ability of EITE sources to implement those measures if it would render them uncompetitive in the marketplace and therefore risk unintended consequences, including GHG leakage. Further, Section II.C.3.a(i)(B)(1) sets forth the Commission’s intent that EITE stationary sources will consider and fully evaluate the possibility of employing CCUS or utilization as a control measure for any emissions unit with direct emissions of 100,000 tons CO₂e and provide that analysis to the Division at least every five-year audit cycle.

The analysis must go beyond a superficial examination of whether other similar facilities have successfully employed these technologies and must look at the state of the technology and whether it can be successfully employed at the source under review. If it is technically possible to employ these technologies, they must then be evaluated for other factors such as cost-effectiveness and other direct or indirect impacts, like competitiveness in the marketplace.

The intent of this assessment is to objectively and rationally uncover the strengths and weaknesses of CCUS technology at the facility. The assessment should include economic and technical issues associated with carbon capture technology at the specific facility. A feasibility analysis is expressly not required by this rule, but if a facility has conducted a full feasibility analysis within the last 5 years, that information will satisfy the requirement to provide the information in Section II.C.3.a.(i)(B)(1). In subsequent audit cycles, these reports can be updated instead of recreated. In the interest of regulatory efficiency, as an alternative to conducting an Energy BMP audit under Sections II.C.2. and 3, the EITE stationary source will be determined to have conducted a qualifying audit and be currently employing Energy BMPs if the stationary source is certified under the EPA Energy Star Program or is registered to the ISO 50001 - standard for energy management. This alternative pathway was adopted because the underlying programs examine and audit the facility's performance as a whole, and are equally or more rigorous and thorough than the Energy BMP audit contemplated in § 25-7-105(1)(e)(IX)(A), C.R.S. The Federal Energy Star Program is a widely accepted "benchmarking" program managed by EPA and requires annual certification that the facility is sufficiently efficient (reaches a score of 75 or higher out of 100).

More information available at: https://www.energystar.gov/industrial_plants/earn-recognition/plant-certification

ISO 50001 is the internationally recognized energy management system developed by the International Organization for Standards which requires certification every 3 years.

More information available at: https://www.iso.org/iso/iso_50001_energy_management_systems.pdf

Both certifications are appropriate alternative means of demonstrating Energy BMP employment as they require rigorous evaluation of an industrial facility's energy performance and the employment of best available energy practices through either benchmarking in the case of Energy Star or certification to a specific set of requirements for holistic energy management of the facility, in the case of ISO 50001. If the EITE stationary source is not certified under EPA's Energy Star Program or ISO 50001, the EITE stationary source must conduct an Energy BMP audit pursuant to Section II.C.2. and 3.

The GHG BAECT and Energy BMP Recommendation

As described in Section II.C.3.a.(iii), the auditor's recommendation must include: (1) The auditor's recommendation on the most effective technology or strategy or suite of control technologies as determined under Section II.C.3.a.; (2) a list of all control measures with levelized costs less than or equal to \$0 as determined in Section II.C.3.a.(i)(D); and (3) recommendations on options that provide greater pollution reduction co-benefits to communities surrounding the EITE stationary source.

The GHG BAECT and Energy BMP recommendation in the audit report serves as the foundation for the GHG BAECT and Energy BMP determination to be made by the Division and ultimately finalized by the Commission. This recommendation consists of the technologies identified through the audit process that would be considered "best" to control and reduce GHG emissions from the emissions units included in the audit scope. The recommendation is documented in the audit report. The GHG BAECT recommendation may be a single technology or reduction measure for the emission unit or a suite of technologies or measures if multiple measures are able to achieve greater reductions and are the same or similar cost-effectiveness.

This approach creates potential for a greater reduction at the same or similar cost to the EITE stationary source, as well as opportunities for increased consideration and inclusion of technologies or measures that have significant co-benefits. The energy BMP recommendation will be issued as a list of the most effective energy efficiency measures for the EITE stationary source for the energy consumption sources included in the audit scope. Both the GHG BAECT and energy BMP recommendations will include a list of measures that have a levelized cost of equal to or less than \$0 for the Division's consideration.

The GHG BAECT and Energy BMP recommendation will document the total cumulative GHG emissions intensity rate for all emission units included in the audit scope. This will be calculated by identifying each emissions unit's annual GHG emissions per final product from the EITE source then summing all audited emission unit intensity rates for a total GHG BAECT and Energy BMP GHG emissions intensity rate for the EITE stationary source. If any energy efficiency measure achieves additional GHG reductions for the facility it shall be included in the GHG BAECT emission intensity rate per final product of the EITE stationary source.

The audit report must also show the calculations and recommendation for the total mass emissions from emission units that were not included in the audit scope. This number is integral to the annual emissions limit calculation for each EITE stationary source because the emissions limitation must include all emissions from the facility.

The GHG BAECT and Energy BMP Intensity Rate Determination

The Division will analyze and review the audit report's GHG BAECT and Energy BMP intensity rate recommendations and associated analyses for all emission units included in the audit scope and make a final recommendation to the Commission to establish the GHG BAECT and Energy BMP GHG emissions intensity rate for the EITE stationary source. The Commission may ask questions, require additional information, or grant approval to the Division-determined rate for the facility. The Division will strive to incorporate these briefings into existing briefings or proceedings before the Commission.

After receiving the audit report, the Division's determination will be based on the audit recommendation and the Division may select either the top control identified or a suite of controls that meet certain, enumerated criteria. The Division's GHG BAECT and Energy BMPs determination for each emissions unit or energy-consuming source within the audit scope will include: (1) The specific control or suite of controls selected as GHG BAECT (Section II.d.1.b.) and Energy BMPs (Section II.D.1.c.); (2) the final GHG BAECT and energy BMP intensity rate for the EITE stationary source (Section II.D.1.b.(iv)); and (3), in addition to any controls selected as GHG BAECT and/or Energy BMPs, all controls found to have a net levelized neutral cost over five (5) years from operational date, unless doing so would impose an unreasonable burden to the EITE stationary source as determined by the Division and based upon findings in the audit report (Sections II.D.1.b.(ii) and II.d.1.c.(ii)).

The GHG BAECT and Energy BMP intensity rate is determined from the GHG BAECT analysis for each piece of GHG emitting equipment to calculate a cumulative GHG BAECT and energy BMP Intensity Rate for the facility. This GHG BAECT and energy BMP intensity rate represents a single, cumulative rate for all GHG emitting equipment included in the audit scope, for both GHG BAECT and energy BMPs. This intensity rate does not include any GHG emitting equipment not included in the audit scope.

The calculation is as follows:

GHG BAECT & Energy BMP Intensity Rate Determination = Σ (CO₂e per tons of facility product for each emission unit in audit scope)

Non-GHG BAECT Emissions Calculation

During the audit, the mass total of GHG emissions from the GHG emitting equipment that was not included in the audit scope, or “Non-GHG BAECT emissions,” shall be determined. Non-GHG BAECT emissions are calculated by subtracting the reported emissions from units evaluated for GHG BAECT from the facility annual emissions at the time of the first audit. Once reviewed and approved by the Division and Commission, this mass total will remain fixed. This number will be used in calculating the annual emissions limitation for each EITE stationary source.

The calculation is as follows:

Non-GHG BAECT Emissions = Total direct emissions from the most recent year reported -
(reported emissions from the units evaluated for GHG BAECT)

Annual Emissions Limitation Determination

The annual emissions limit must consider all GHG emissions from the EITE stationary source. The direct GHG emitting units that were not included in the audit scope must be included in calculating the limit. Therefore, the GHG BAECT and energy BMP intensity rate determined by the Division must be converted to mass by using the year's production total. *This is done by multiplying the units of production for the EITE stationary source for the year, by the GHG BAECT and energy BMP intensity rate. This results in a mass emissions total which represents the mass emissions from the facility operating all GHG BAECT and energy BMPs during that year. The Non-GHG BAECT emissions are then added to this total, resulting in a mass-based emission total for the EITE stationary source. This is then multiplied by 95% in order to affect the 5% reduction required by statute, the end result of which is the EITE stationary source's "annual emission limit" for the year under review.* Calculation of and compliance with this annual emissions limitation is required beginning in the third year after the audit year, onward. The five percent reduction must be achieved in addition to any subsequent lower GHG BAECT and Energy BMP intensity level determined for the EITE stationary source.

The calculation is as follows:

Annual Emission Limit for Compliance Year in tons CO₂e = (((GHG BAECT & Energy BMP Intensity Rate Determination) * (Compliance Year Facility Product)) + (Non-GHG BAECT Determination)) * 0.95

Compliance Year Facility Product = tons of final product reported to Division by EITE Stationary Source

This calculation links the mass based number to total production and is appropriate for EITE industries. This is a unique set of industries, where there is the potential for “leakage”- that is, for production to move to other locations outside of Colorado, and because of this, state statute specifically treats EITE industries distinctly from other GHG emitting industries. This mass based approach to production may not be appropriate for the industrial sector as a whole or other sectors. This approach gives effect to § 25-7-105(1)(e)(XIII)(B), C.R.S., because it would be consistent with the requirements of § 25-7-105(1)(e)(IX) and would affect a 5% reduction in GHG emissions “associated with [EITE] sources that currently employ [GHG BAECT] and [Energy BMPs], as determined by the [C]ommission....” Pertinent to these reduction requirements, where an EITE source is employing GHG BAECT and Energy BMPs, § 25-7-105(1)(e)(IX) limits the Commission’s ability to “impose a direct non-administrative cost on the source directly associated with at least [95%] of the source’s [GHG] emissions attributable to manufacturing a good in the state[.]” § 25-7-105(1)(e)(IX)(A), C.R.S. Thus, in order to give effect to this limitation and the emissions reduction required in § 25-7-105(1)(e)(XIII)(B), the 5% reduction requirements must apply to GHG emissions “associated with the [EITE] sources” and “attributable to manufacturing a good in the state.”

Use of the emissions limitation approach set forth in GEMM for EITE sources accomplishes this suite of legislative directives. It requires EITE sources to reduce GHG emissions “attributable to manufacturing a good in the state by 5%.” It does not impose non-administrative costs on the other 95% of GHG emissions from the source. This provides the EITE sources flexibility to mitigate the costs of the required reductions and provides an incentive to improve efficiency and reduce emissions.

Importantly, it accomplishes all this while allowing EITE sources to adjust production levels based on market forces and meet demand with an annual mass emissions limitation at a GHG intensity of 5% below that accomplished through the employment of GHG BAECT and Energy BMPs without requiring the source to reduce production to accomplish these reductions. This balances the need for substantial and lasting GHG emissions reductions from EITE sources while recognizing the treatment afforded these particular sources by the General Assembly.

Annual Emission Limit Compliance

Pursuant to Sections II.E.1.b. and II.F.2., EITE stationary sources are required no later than the third year after each audit to achieve and maintain an additional mass-based five percent GHG emission reduction below emissions that would be achieved through the employment of GHG BAECT and Energy BMPs, unless an interim emission rate is established under Section II.E.1.c. To demonstrate compliance, EITE stationary sources must submit an annual compliance report with the information set forth in Section II.G.1., including the previous year’s GHG emissions; units of product produced; and the GHG BAECT and energy BMP intensity rate determination. The EITE stationary sources must also maintain records for 10 years.

Compliance Pathways

To give effect to the legislative directive § 25-7-105(1)(e)(XIII)(B), C.R.S., and in accordance with the guidance concerning implementing rules in §§ 25-7-105(1)(e)(II), (V), and (VI), C.R.S., in Section II.F.1.c. the Commission provides three pathways to achieve the required emission intensity rate. This includes any one or combination of: direct on-site reductions (Section II.F.1.a.(i)); surrendering reduction credits created by GHG reductions at other regulated sources (Section II.F.1.a.(ii)); and utilization of retail distributed generation or net metering renewable projects that reduce GHG emissions from the facilities’ energy use for which RECs have been retired (Section II.F.1.a.(iii)). For the utilization of renewable energy, the RECs retired must be from Colorado and retired in the year generated. Regardless of the compliance pathway utilized, the EITE stationary source must assure that any significant co-benefits are achieved at the EITE stationary source pursuant to Section II.F.1.b.

Utilization of Renewable Energy Credits for compliance cannot exceed the annual generation of the distributed generation system and cannot be used for compliance with the Annual Emissions Limitation for more than the required 5% reduction, nor can they be counted toward the 20% emissions reduction required to exempt the source from further 5% reduction requirements, as those are to be based on direct emissions reductions from the facility.

The megawatt hour (MWh) of distributed generation in avoided GHG emission value shall be based on the relevant electricity provider’s reported annual system emissions, calculated in a manner consistent with the Division’s AQCC-approved Clean Energy Plan guidance, and be adjusted annually to account for the changing emissions profile of the generation fleet of the source’s electricity supplier. This is because a system MWh avoided today may be more emissions intensive than a system MWh avoided in the future.

The Commission recognizes that large capital-intensive emissions reduction projects such as large-scale carbon capture, utilization, and storage may take an extended time to complete. The Commission directs staff to develop a proposal to allow an alternative compliance pathway with an extended timeline, with sufficient guardrails to ensure that cumulative GHG emissions reductions will be greater than those achievable with short-term measures, and that communities near the facility will realize appropriate co-benefits.

Points of Compliance

There are a number of discrete requirements with which the covered sources must demonstrate compliance over the course of each audit cycle. These include:

- Submission of the Audit Plan at least 120 days prior to commencing the audit (Section II.C.2.);
- Submission of the Audit Report no later than December 31, 2022 and every five years thereafter (Section II.C.1.a.);
- Submission and execution of the Compliance action plan within 120 days of the Commission's approval of the GHG BAECT and Energy BMP determination (Section II.F.1.);
- Compliance with annual emission limits established in the GHG BAECT and Energy BMP determination by the third year after each audit year (Sections II.E.1.b. and II.F.2); and
- Submission of annual compliance certification by May 1 beginning in 2026 (Section II.G.1).

Mechanisms for Assuring Compliance

GEMM is generally enforceable as a provision of part 1 of the Act, see § 25-7-105(1)(e)(IX) (requiring EITE stationary sources to conduct GHG BAECT and Energy BMP audits), and as an emission control regulation promulgated by the Commission. See § 25-7-115(1)(a), C.R.S. Accordingly, noncompliance with GEMM is subject to the injunctive and civil penalty authority of the Division and Commission. See §§ 25-7-121 and -122, C.R.S. In addition to these general authorities, f § II.K.1.a. provides an alternative means of addressing noncompliance with the emission reduction requirements of GEMM where the EITE stationary source is utilizing GHG credits to comply. Section II.K.1.a. provides the source an opportunity to rectify any shortcomings in emission reduction requirements under GEMM by surrendering three GHG credits for every ton emitted by the source in excess of its annual emission limit.

Co-benefits

The treatment of co-benefits in the evaluation and determination of co-benefits in the audit and subsequent effects on the Section II.F. compliance pathways and is affected through three mechanisms:

1. In Section II.C.3.a.(i)(G) for BAECT and Section II.C.3.a.(ii)(A)(7) for BMPs, the auditor must evaluate and document expected air quality co-benefits of any BAECT or BMP measure assessed;
2. In Section II.C.3.a.(iii)(A)(4) for BAECT and Section II.C.3.a.(iii)(B)(3) for BMPs, the auditor's recommendation must include as a BAECT and/or BMP the option that provide greater co-benefits to the surrounding communities where the top options are otherwise comparable in terms of cost-effectiveness; and
3. In Section II.F.1.b., where an EITE source complies with the annual emission limit, it must also achieve the co-benefits to the local community where "the measure(s) determined to be GHG BAECT and/or energy BMPs for an emission unit also are anticipated to result in significant co-benefits[.]"

As to this final point, the express intent is to ensure that any localized co-benefits that would be realized through implementation of the specific BAECT or BMPs identified in the audit are still realized for the local community, regardless of which compliance pathway the source utilizes. This is explicitly intended to benefit local communities and particularly those disproportionately impacted by climate change and other air quality issues and is in response to input from communities and local governments.

The quantification of any co-benefits under Sections II.C.3.a.(i)(G) and II.C.3.a.(ii)(A)(7) must include establishment of a baseline for the relevant pollutant(s) and quantification of net co-benefits from the control technologies and strategies below or above the baseline. The baseline shall be determined based on the best available information, including monitored emissions, if available, or reported emissions of the relevant pollutant(s) at the time of the audit.

GHG Credit Accounting and Trading Program for EITE Stationary Sources

§ 25-7-105(1)(f)(I)(C), C.R.S., defines “trading program” as “a commission-adopted regulatory program that allows for regulated sources to meet their greenhouse gas compliance obligations under subsection (1)(e) of this section through the creation, purchase, acquisition, or exchange of, or other commercial-type transaction involving, a GHG credit with other regulated sources.” § 25-7-105(1)(f)(I)(A), C.R.S., defines “regulated source” as “a source of [GHG] that is subject to a rule adopted by the [C]ommission under [§ 25-7-105(1)(e)] that imposes specific and quantifiable [GHG] reduction obligations upon that source or group of sources.”

An accounting and “trading program” for GHG credits will be developed by the Division, which can be utilized by EITE sources covered by GEMM. The trading program will serve as one pathway to complying with GHG emissions limitations and will allow trading of GHG reductions while also ensuring co-benefits of reducing localized harmful air pollutants at the EITE sources. This is achieved by separately determining what reductions in harmful air pollutants will be achieved at the source through the use of GHG BAECT and requiring that these emission reductions are achieved regardless of how the EITE complies to reduce its GHG emissions.

GHG credits issued by the Division will serve as the mechanism for the trading program. These GHG credits must be utilized for the trading program and represent a GHG emission reduction of one metric ton of CO₂e in Colorado, and be real, additional, quantifiable, permanent, verifiable and enforceable. The Act authorizes the Commission to adopt this program through its general rulemaking authority, its authority to adopt implementing rules for GHG pollution reduction, and specific authority to adopt an accounting and trading program as established under House Bill 21-1266. See §§ 25-7-106(1) (granting the Commission “maximum flexibility in developing an effective air quality control program and [the authority to] promulgate such combination of regulations as may be necessary or desirable to carry out that program”); -105(1)(e)(II) (granting the Commission broad authority to adopt implementing rules to affect GHG emissions reductions); -105(1)(e)(V) (implementing rules for GHG reductions should enhance cost-effectiveness and compliance flexibility and transparency around compliance costs); -105(1)(e)(IX)(A) (EITE sources should be provided “a pathway to obtain equivalent lower-cost emission reductions at other regulated sources to satisfy their compliance objectives”); and -105(1)(f) (granting the Commission authority to establish GHG credit trading between “regulated sources,” and specifically to implement § 25-7-105(1)(e)(IX)).

The program accords with the Commission’s authority by allowing EITE sources pathways to accomplish reduction requirements at other regulated sources using an accounting and trading program through which the Division can track emission reductions and trades, prevent double-counting of GHG emission reductions, and identify EITE sources that adversely affect disproportionately impacted communities through emissions of locally harmful air pollutants. Importantly, the accounting and trading program applies only to EITE stationary sources subject to the audit and GHG emission reductions in GEMM.

However, given the authority granted to the Commission in § 25-7-105(1)(f), it is possible that this program may serve as a model for a future, more broadly applicable GHG credit trading for the industrial manufacturing sector. Should that arise, the Commission may consider how EITE sources should be incorporated into the broader program at that time. The current approach recognizes the directive in statute that EITE sources be provided “a pathway to obtain equivalent lower-cost emission reductions at other regulated sources to satisfy their compliance obligations” while also recognizing EITE sources would initially be the only “regulated sources” as that term is defined, and therefore the only sources eligible to participate in the trading program until additional GHG sources in Colorado meet this definition.

Based on these tenets, Section II.I. sets out provisions establishing an accounting system to track GHG reduction credits. These provisions task the Division with establishing the accounting system. An EITE source seeking eligibility to trade credits must first apply and have the Division create a compliance account for the EITE source. This application for an account must be submitted to the Division within 30 days after approval of the EITE source's BAECT and Energy BMP determination. Any EITE source the Division determines adversely affects disproportionately impacted communities through emission of locally harmful air pollutants must be identified by the Division in the accounting system.

Only GHG credits meeting the definition set forth in Section II.B.22 may be traded in the accounting system. Pursuant to this definition a "GHG credit" represents a GHG emission reduction of one metric ton of CO₂e that is real, additional, quantifiable, permanent, verifiable and enforceable. To track such credits, the Division is directed to assign each credit a unique identifier (such as a serial number) and, as further directed in Section II.F.1.a.(ii), only issued in the tracking system after the underlying emission reduction has been demonstrated.

Sections II.G. and II.H. are designed to ensure that EITE stationary sources conduct regular reporting to demonstrate compliance with the audit requirements in Section II.C. and emissions reduction requirements in Section II.F. and maintain all pertinent records. The Commission determined that the audit procedures and compliance requirements set out in Part B, Section II, establish the criteria by which the Commission can determine, on a five-year basis, whether EITE stationary sources are employing GHG BAECT and energy BMP. The Commission has determined the audit process is cost-effective and reasonable to achieve these ends.

Additional Considerations

The following are additional findings of the Commission made in accordance with the Act:

§ 25-7-110.5(5)(b), C.R.S.

As these revisions exceed and may differ from the federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), C.R.S., the Commission determines:

- (I) Any federal requirements that are applicable to this situation with a commentary on those requirements;

EITE Entities are required to report GHG emissions under existing federal regulations. Some EITE stationary sources may be required to conduct a GHG "best available control technology" or "BACT" analysis as part of a Prevention of Significant Deterioration permitting action unrelated to the requirements of these rules. However, there are no current federal regulations requiring these entities to conduct GHG BAECT and energy audits or to reduce GHG emissions as required under this rule.

- (II) Whether the applicable federal requirements are performance-based or technology-based and whether there is any flexibility in those requirements, and if not, why not;

EITE Entities are required to report GHG emissions under existing federal and state regulations. Some EITE stationary sources may be required to conduct a GHG "best available control technology" or "BACT" analysis as part of a Prevention of Significant Deterioration permitting action unrelated to the requirements of these rules. Those BACT evaluations are technology-based. However, there are no current federal regulations requiring these entities to conduct GHG BAECT and energy audits or reduce GHG emissions as required under this rule.

- (III) Whether the applicable federal requirements specifically address the issues that are of concern to Colorado and whether data or information that would reasonably reflect Colorado's concern and situation was considered in the federal process that established the federal requirements;

Federal BACT analyses under PSD permitting are separate and distinct from the GHG BAECT analysis adopted in this Part B, Section II. in both its scope and purpose. Under federal PSD permitting, GHG emissions cannot alone trigger a BACT analysis, though the permitting authority may evaluate GHG controls for “anyway” sources that trigger PSD permitting requirements by exceeding other criteria pollutant thresholds. *See Air Regul. Grp. v. E.P.A.*, 573 U.S. 302, 333–34 (2014). This Part B, Section II. in contrast is adopted pursuant to specific legislative directive to evaluate GHG BAECT and Energy BMPs; there are no applicable federal requirements in this regard.

- (IV) Whether the proposed requirement will improve the ability of the regulated community to comply in a more cost-effective way by clarifying confusing or potentially conflicting requirements (within or cross-media), increasing certainty, or preventing or reducing the need for costly retrofit to meet more stringent requirements later;

This section gives meaningful effect to §§ 25-7-105(1)(e)(IX) and (XIII), C.R.S., and provides regulated entities flexibility to identify and cost-effectively employ emissions control technologies BAECT and energy efficiency measures to reduce GHG emissions so long as they constitute GHG BAECT and Energy BMPs. Regulated entities that demonstrate effective employment of BAECT and Energy BMPs are afforded certainty with respect to direct non-administrative costs associated with ninety-five percent of the source’s GHG emissions and control over the means of otherwise reducing GHG emissions to comply with § 25-7-105(1)(e)(XIII)(B), C.R.S.

- (V) Whether there is a timing issue which might justify changing the time frame for implementation of federal requirements;

There are no applicable federal requirements that create any timing issues. Part B, Section II. allows regulated entities a reasonable time to comply with the audit, any resulting compliance action plan, and GHG emission reduction requirements and allows opportunities for alternative compliance.

- (VI) Whether the proposed requirement will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth;

Part B, Section II. affords regulated entities significant flexibility for meeting GHG emission reduction requirements. Furthermore, EITE entities are afforded the ability to affect required reductions through alternative compliance measures where needed. As such, regulated entities are afforded a reasonable margin for accommodation of uncertainty and future growth. The three pathways provided for EITE stationary sources to accomplish the 5% emissions reductions in Sections II.E. and II.F. and as required under § 25-7-105(1)(e)(XII)(B), C.R.S., provide affected sources flexibility to accommodate uncertainty and future growth.

- (VII) Whether the proposed requirement establishes or maintains reasonable equity in the requirements for various sources;

All qualifying EITE entities are equally subject to the audit and action plan requirements and the audits account for the specifics of the EITE stationary source in question. Likewise, all regulated entities are afforded opportunities to select and apply BAECT and energy BMPs that work for the specific source and account for environmental, economic, and energy concerns. Furthermore, the five percent GHG emission reduction requirements in Sections II.E. and II.F. are equally applied to all qualifying EITE stationary sources. Where available, opportunities to maximize co-benefits to communities near or around the stationary source should be prioritized.

- (VIII) Whether others would face increased costs if a more stringent rule is not enacted;

The General Assembly has acknowledged that climate change impacts Colorado's economy and directed that GHG emissions should be reduced across all sectors of our economy. Colorado has established specific GHG reduction goals. Reductions not achieved in one sector will require measures in other sectors of the economy to achieve the state's GHG reduction goals.

However, the General Assembly further provided requirements that energy-intensive and trade-exposed entities demonstrate use of BAECT and energy BMPs through an audit process and limited the Commission's ability to impose additional reductions on at least ninety-five percent of the source's GHG emissions where such measures are effectively employed. See § 25-7-105(1)(e)(IX), C.R.S. With respect to the 5 percent emission reductions required of EITE entities employing GHG BAECT and energy BMPs under § 25-7-105(1)(e)(XIII)(B), C.R.S., any emission reductions not timely realized from these entities would, in turn, require greater emission reductions from other sources in the industrial and manufacturing sector in order to achieve the twenty percent sector-wide requirements by 2030 set forth in § 25-7-105(1)(e)(XIII)(A), C.R.S.

- (IX) Whether the proposed requirement includes procedural, reporting, or monitoring requirements that are different from applicable federal requirements and, if so, why and what the "compelling reason" is for different procedural, reporting, or monitoring requirements;

Part B, Section II. gives effect to the General Assembly's adoption of 25-7-105(1)(e)(IX), C.R.S., which includes a requirement for energy-intensive trade-exposed entities to execute energy and emission control audits that are not required under federal regulations. This is a compelling reason, as these audits will inform the state's strategies and future regulations to accomplish the statewide GHG pollution reduction goals and address the impacts of climate change set forth in § 25-7-102(2), C.R.S. and sector-specific emission reductions under § 25-7-105(1)(e)(XIII), C.R.S.

- (X) Whether demonstrated technology is available to comply with the proposed requirement;

Part B, Section II. does not require the use of any specific technology but instead serves as a mechanism to evaluate the control technologies and energy efficiency practices regulated entities are employing and to determine the effectiveness of those measures already in use. The GHG BAECT and Energy BMP audits are used to conduct this evaluation and must include analyses of, but not necessarily implementation of, transformative technologies. All GHG BAECT and energy BMP determinations will be based on demonstrated and available technologies.

- (XI) Whether the proposed requirement will contribute to the prevention of pollution or address a potential problem and represent a more cost-effective environmental gain;

This rule will enable the Commission to determine whether EITE stationary sources are employing GHG BAECT and Energy BMPs to effectively minimize GHG emissions from regulated facilities. EITE sources will be required to comply with the annual emission limits established pursuant to this audit process. The GHG emissions reductions from this rule are expected to help Colorado achieve the statewide GHG pollution reduction goals in § 25-7-102(2)(g), C.R.S., and the sector-specific GHG emission reductions set forth in § 25-7-105(1)(e)(XIII), C.R.S. Anticipated reductions in co-pollutants are expected to have positive health benefits for the people of Colorado.

- (XII) Whether an alternative rule, including a no-action alternative, would address the required standard.

This rule implements the statutory requirements of §§ 25-7-105(1)(e)(IX) and (XIII)(B), C.R.S. Alternatives exist for how to accomplish these requirements, including different emission thresholds for qualifying entities, different standards for evaluating GHG BAECT and Energy BMPs, and the provision of no or differing means of alternative compliance as well as different timing requirements for emission reductions. The Commission determined that Part B, Section II. appropriately gives effect to the statutory requirements and is consistent with the statewide and sector-specific GHG pollution reduction goals. A no-action alternative is not available under § 25-7-105(1)(e), C.R.S.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

- (I) These rules are based on reasonably available, validated, reviewed, and sound scientific methodologies and all validated, reviewed, and sound scientific methodologies and information made available by interested parties has been considered.
- (II) Evidence in the record supports the finding that the rule shall result in a demonstrable reduction in GHG pollution and co-pollutants and will enable the Commission to satisfy the requirements of §§ 25-7-102, -105(1)(e), -106, and/or -109, C.R.S., as applicable.
- (III) Evidence in the record supports the finding that the rule shall bring about reductions in risks to human health and the environment that will justify the costs to government, the regulated community, and to the public to implement and comply with the rule.
- (IV) The rules are the most cost-effective to achieve the necessary and desired results and reduction in air pollution.
- (V) The rule will maximize the air quality benefits of regulation in the most cost-effective manner.

II. Adopted: July 21, 2022

(Removed from Regulation Number 22 and placed in Regulation Number 27 April 20, 2023)

Revisions to Regulation Number 22, Part B, Sections II.B.19. and II.B.25.

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-103, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, §§ 25-7-110 and 25-7-110.5, C.R.S., and the Air Quality Control Commission's (Commission) Procedural Rules.

Basis

The Commission has incorporated by reference Regulation Number 22, Greenhouse Gas Emissions and Energy Management for Manufacturing in Colorado, Part B, Section II.B.19. and Part B, Section II.B.25. These sections shall be modified for the limited purpose of removing the incorporation by reference sentence in Part B, Section II.B.19. and Part B, Section II.B.25. This correction does not change or alter the requirements of the existing rule.

Specific Statutory Authority

The Colorado Air Pollution Prevention and Control Act, §§ 25-7-105(1)(b) and 25-7-109, C.R.S. authorize the Commission to adopt emission control regulations, including emission control regulations relating to new stationary sources, for the development of an effective air quality control program.

Purpose

Updating citation references of 40 C.F.R. Part 60 allows the Division to implement and enforce the Emission Guidelines and Compliance Times for applicable source categories. Adoption of the rules will not impose additional requirements upon sources beyond the minimum required by federal law and may benefit the regulated community by providing sources with up-to-date information and regulatory certainty.

III. Adopted: April 20, 2023

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

To improve the readability and usability of Regulation Number 7 and Regulation Number 22, the Commission adopted revisions restructuring and reorganizing the parts and sections.

Specific Statutory Authority

The Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq. (the State Air Act or the Act), specifically § 25-7-103.3, directs rule-making agencies, such as the Commission, to review their rules and consider whether the rule is necessary; whether the rule overlaps or duplicates other rules of the agency or with other federal, state, or local government rules; whether the rule is written in plain language and is easy to understand; whether the rule has achieved the desired intent and whether more or less regulation is necessary; whether the rule can be amended to give more flexibility, reduce regulatory burdens, or reduce unnecessary paperwork or steps while maintaining its benefits; whether the rule is implemented in an efficient and effective manner, including the requirements for the issuance of permits and licenses; whether a cost-benefit analysis was performed by the applicable rule-making agency; and whether the rule is adequate for the protection of the safety, health, and welfare of the state or its residents. Based on this review, the rule-making agency will determine whether the existing rules should be continued in their current form, amended, or repealed.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 27.

The Commission reorganized Regulation Number 7 into four regulations: Part B became Regulation Number 24; Part C became Regulation Number 25; Part D remained in Regulation Number 7; and Part E became Regulation Number 26. The upstream oil and gas intensity and midstream combustion program provisions currently in Regulation Number 22 moved to Regulation Number 7. The manufacturing sector greenhouse gas provisions in Regulation Number 22 became a new Regulation Number 27.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

These revisions are administrative in nature and, therefore, do not exceed or differ from the requirement of the federal act or rules. Therefore, § 25-7-110.5(5)(a) does not apply.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of greenhouse gas and VOC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.
- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.
- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

IV. Adopted: October 20, 2023

Revisions to and reorganization of Regulation Number 27.

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-103(4), C.R.S., the Colorado Air Pollution Prevention and Control Act, §§ 25-7-110 and -110.5, C.R.S., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

In House Bill 19-1261, now codified in part at §§ 25-7-102(2) and -105(1)(e), C.R.S., the General Assembly declared that "[c]limate change adversely affects Colorado's economy, air quality and public health, ecosystems, natural resources, and quality of life[.]" acknowledged that "Colorado is already experiencing harmful climate impacts[.]" and acknowledged that "[m]any of these impacts disproportionately affect" certain disadvantaged communities. § 25-7-102(2), C.R.S. The General Assembly also recognized that "[b]y reducing greenhouse gas pollution, Colorado will also reduce other harmful air pollutants, which will, in turn, improve public health, reduce health care costs, improve air quality, and help sustain the environment." § 25-7-102(2)(d), C.R.S. Consequently, the General Assembly updated Colorado's statewide greenhouse gas (GHG) pollution reduction goals so as to achieve a 26% reduction of statewide GHG by 2025; 50% reduction by 2030; 65% reduction by 2035; 75% reduction by 2040; 90% reduction by 2045; and 100% reduction by 2050, as compared to 2005 levels. Senate Bill 2023-16, 74th Gen. Assemb., 1st Reg. Sess., Sec. 8 (Colo. 2023) (amending § 25-7-102(2)(g), C.R.S.).

In 2021, Colorado's legislature adopted House Bill 21-1266 (the Environmental Justice Act), now codified in part at § 25-7-105(1)(e), C.R.S. This provision grants the Commission broad authority to regulate GHG emissions to accomplish the goals established in § 25-7-102(2)(g), C.R.S. It directs the Commission to adopt rules that reduce statewide GHG emissions from the industrial and manufacturing sector by 2030 by at least 20% below the 2015 baseline, according to the Colorado State GHG Inventory. The rules must include protections for disproportionately impacted communities and prioritize emission reductions that will reduce emissions of co-pollutants that adversely affect disproportionately impacted communities. The rules must also be designed to accelerate near-term reductions and secure meaningful emission reductions from the industrial and manufacturing sector to be realized no later than September 30, 2024. The Commission adopted requirements for certain energy-intensive, trade-exposed (EITE) manufacturing stationary sources in October 2021 (GEMM 1) and then adopted requirements for additional stationary sources in the manufacturing sector in September 2023 (GEMM 2).

The Commission also restructured the regulation into the following Parts: Part A, General Provisions (fka Part A, Sections I., II., X., and XI.); Part B, GEMM 2 Facility Requirements (new); Part C, Energy-Intensive Trade-Exposed Stationary Source Requirements (fka Part B, Sections III. through IX.); and Part D, Greenhouse Gas Credit Trading Requirements (fka Part B, Section IX.).

Specific Statutory Authority

The Act, specifically § 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102, C.R.S., and that are necessary for the proper implementation and administration of the Act.

Section 25-7-105(1)(e), C.R.S., authorizes the Commission to promulgate implementing rules and regulations consistent with the statewide GHG pollution reduction goals in § 25-7-102(2)(g), C.R.S. In adopting GHG abatement strategies and implementing rules, the Commission is authorized to take into account other relevant laws and rules, as well as voluntary actions of local communities and the private sector, to enhance efficiency and cost-effectiveness, and to solicit input from other state agencies and stakeholders on the advantages of different statewide GHG pollution mitigation measures. § 25-7-105(1)(e)(II) and (IV), C.R.S.

Implementing rules may include regulatory strategies that incentivize development of renewable resources and "enhance cost-effectiveness, compliance flexibility, and transparency around compliance costs." § 25-7-105(1)(e)(V), C.R.S. Further, in promulgating such implementing rules, the Commission is to consider many factors, including, but not limited to: health, environmental, and air quality benefits and costs; the costs of and time necessary for compliance; the relative contribution of each source or source category to statewide GHG pollution; equitable distribution of the benefits of compliance; issues related to the beneficial use of electricity to reduce GHG emissions; and whether greater or more cost-effective emission reductions are available through program design. § 25-7-105(1)(e)(VI), C.R.S.

§ 25-7-106, C.R.S., provides the Commission "maximum flexibility in developing an effective air quality program and [promulgating] such [a] combination of regulations as may be necessary or desirable to carry out that program." Section 25-7-109(1), C.R.S., authorizes the Commission to adopt and promulgate emission control regulations that require the use of effective practical air pollution controls for each type of facility, process, or activity which produces or might produce significant emissions of air pollutants. An "emission control regulation" may include "any regulation which by its terms is applicable to a specified type of facility, process, or activity for the purpose of controlling the extent, degree, or nature of pollution emitted from such type of facility, process, or activity. . . ." § 25-7-103(11), C.R.S. Emission control regulations may pertain to any chemical compound including GHG pollution. See § 25-7-109(2)(c), C.R.S.

§§ 25-7-105(1)(b) and 25-7-109, C.R.S. authorize the Commission to adopt emission control regulations, including emission control regulations relating to new stationary sources, for the development of an effective air quality control program.

Purpose

In 2021, the Commission adopted Regulation Number 27, which established requirements for certain EITE manufacturing stationary sources (GEMM 1). The Commission then revised Regulation Number 27 to require GHG emission reductions from additional stationary sources that emit equal to or greater than 25,000 metric tons of CO₂e (GEMM 2). The Commission also aligned the applicability threshold for GEMM 1 facilities with the applicability threshold for GEMM 2 facilities, from 50,000 metric tons of CO₂e to 25,000 metric tons of CO₂e, to allow for consistency in the regulatory approach.

GEMM 2 Facilities

As of the effective date of the rule, the GEMM 2 requirements set out in Part B impact 18 facilities (covered GEMM 2 facilities) and are intended to achieve a reduction in the cumulative direct GHG emissions from the covered GEMM 2 facilities of almost 12% by 2024 and 20% by 2030, relative to 2015 levels. The rule also applies to any other manufacturing facilities that exist as of the effective date of the rule and emit equal to or greater than 25,000 metric tons of CO₂e in any year following the effective date of the rule. Apart from facilities subject to GEMM 1, the facilities covered by this rule are the manufacturing sources with the largest annual direct GHG emissions in the State. The facilities are defined as manufacturing operations by the North American Industrial Coding System, and have processes at their facilities that include the mechanical, physical, or chemical transformation of materials, substances, or components into new products. With one exception, the 18 covered GEMM 2 facilities are not considered EITE sources.

Covered GEMM 2 Facilities GHG Reductions

The Commission adopted requirements for covered GEMM 2 facilities to achieve onsite GHG reductions from their baseline emissions through technically feasible and economically reasonable reduction measures such as equipment upgrades, increasing efficiency, and installation of additional controls. Onsite carbon capture and storage (CCS) is another potential reduction measure pending the Division establishing or adopting by reference a standardized CCS protocol(s).

For covered GEMM 2 facilities, baseline emissions were established as the higher of the facility's 2021 or 2022 GHG emissions, subject to certain revisions to correct previous inaccuracies and certain production capacity-related adjustments. Facilities that had increased production capacity between 2015 and 2021 by more than 30%, but which had not yet realized all of the additional production capacity through actual production, were eligible for baseline adjustments as set forth below. The adjustment depended on the size of the production increase, the year(s) in which the expansion occurred, production levels after the expansion, and the GHG emissions increase estimated from the expansion. All facilities were granted 75% of their requested GHG emissions increase, except for facilities which, as of rule adoption, have already achieved a 20% reduction compared to 2015, which were granted 100% of their requested GHG emissions increase.

Although subject to change to correct prior data or reporting inaccuracies, as of the date of rule adoption, the higher of 2021 or 2022 reported GHG emissions and the baselines of the covered GEMM 2 facilities are shown below in Table 1. The figures in Table 1 rely on the 100-year global warming potentials published in the Fifth Assessment Report (AR5) from the Intergovernmental Panel on Climate Change Working Group 1 (see IPCC, Fifth Assessment Report, https://www.ipcc.ch/site/assets/uploads/2018/02/WG1AR5_Chapter08_FINAL.pdf).

TABLE 1

Facility Name	GEMM 2 facility GHG emissions: Higher of 2021 or 2022 year (metric tons of CO ₂ e using AR5 GWP values)	GEMM 2 facility GHG baseline emissions (metric tons of CO ₂ e using AR5 GWP values)
American Gypsum Company	75,047	75,047
Anheuser Busch Inc., Fort Collins Brewery	43,710	43,710
Avago Technologies	125,339	125,339
Carestream Health, Inc.	34,894	34,894
Cargill Meat Solutions	39,588	39,588
Front Range Energy	41,312	60,369
Golden Aluminum Inc.	26,759	26,759
JBS Swift Beef Company - Greeley Plant	171,101	171,101
Leprino Foods, Greeley	97,816	132,878
Microchip Technology	168,907	168,907
Molson Coors USA LLC - Golden Brewery	234,938	234,938
Natural Soda	49,309	56,227
Owen-Brockway Glass Container Plant	116,002	116,002
Rocky Mountain Bottle Company	76,684	76,684
Sterling Ethanol, LLC	56,370	56,370
Suncor Energy USA, Commerce City	951,898	951,898
Western Sugar Cooperative	81,981	109,141
Yuma Ethanol, LLC	55,500	55,500

In establishing the GHG reduction criteria set forth in Part B, Section I.A., the Commission considered each covered GEMM 2 facility's GHG reductions since 2015 and their contribution in terms of percent GHG emissions towards the cumulative emissions from the 18 covered GEMM 2 facilities. The resulting percent reduction requirements are reflected in Section I.A. of Part B. The primary driver for requiring an individual facility to reduce a certain amount was how much the facility had already reduced GHG emissions since 2015. The reductions achieved by each facility between 2015 and the facility's higher reported emissions in 2021/2022 varied widely; some covered GEMM 2 facilities had reduced well over 20%, while others had increased emissions by over 100%, and many of the facilities were somewhere in between.

The rule assigned tiered reduction requirements based on the facilities' achieved reductions in an equitable manner. As reflected in Tables 1 through 4 of Part B, those facilities that had achieved significant GHG emission reductions since 2015 were assigned a lesser 2030 reduction obligation than those that had reduced emissions by less than 20% or had increased emissions. This struck a balance that will enable the group as a whole to achieve a 20% reduction by 2030, while requiring all facilities to be on a downward GHG emission trajectory. As reflected in Table 5, facilities that are the larger emitters in the group were assigned an additional percentage of reduction required, depending on their contribution to the group's cumulative emissions. This approach was justified because a minimal decrease in reduction obligation for the higher emitters would have created a significant increase on the required reductions of smaller emitting facilities.

Further, facilities with larger emissions tend to have more opportunities to reduce GHG emissions than smaller facilities. Oftentimes facilities with larger emissions have multiple, varied emission sources for which the facility can analyze for reduction opportunities through, for example, efficiencies and technology improvements. The reductions in Section I.A. were calculated based on the higher of each facility's 2021/2022 emissions and did not consider any baseline adjustment granted to any facility.

Additionally, as set forth in Part B, Sections I.A.1. through I.A.4., the Commission determined that each covered GEMM 2 facility would be limited to its baseline emissions (subject to an exception described below) or required to reduce an additional amount, between 1.25% and 1.75% below its baseline, beginning in 2024. Again, the degree of any reduction required depended on how much, if at all, the facility had already reduced emissions since 2015. The purpose of this requirement was to satisfy the Commission's statutory obligation to design GHG reduction rules that "accelerate near-term reductions" and "secure meaningful reductions" from the industrial and manufacturing sector "to be realized beginning no later than September 30, 2024." § 25-7-105(1)(e)(XIII), C.R.S.

Applying the requirements established in Part B, Section I to the covered GEMM 2 facilities, the Commission determined the resulting emission reduction obligations as shown in Table 2.

TABLE 2

Facility Name	2024–2029 GHG emissions reduction requirement vs. facility baseline (%)	2030 GHG emissions reduction requirement vs. facility baseline (%)
American Gypsum Company	1.75%	12.5%
Anheuser Busch Inc., Fort Collins Brewery	1.25%	7%
Avago Technologies	0.00%	4%
Carestream Health, Inc.	1.75%	12.5%
Cargill Meat Solutions	1.75%	12.5%
Front Range Energy	1.75%	12.5%
Golden Aluminum Inc.	1.50%	8%
JBS Swift Beef Company, Greeley	1.75%	15.5%
Leprino Foods, Greeley	1.75%	12.5%
Microchip Technology	0.00%	4%
Molson Coors USA LLC, Golden	0.00%	4%
Natural Soda	1.50%	8%
Owen-Brockway Glass Container Plant	1.75%	12.5%
Rocky Mountain Bottle Company	1.25%	7%
Sterling Ethanol, LLC	1.75%	12.5%
Suncor Energy USA, Commerce City	1.50%	14%
Western Sugar Cooperative	0.00%	1%
Yuma Ethanol	1.75%	12.5%

Notwithstanding the above, the Commission recognized that some facilities have made significant reductions in mass-based, direct GHG emissions since 2015. Considering this, and to provide near-term flexibility for those specific facilities, the Commission allowed facilities that have already reduced emissions by 20% or more since 2015 to emit up to 75% of the individual facility's reported 2015 emissions through the year 2025, as long as the facility returns to its GEMM 2 facility baseline emissions in 2026. If a facility continues to emit above its GEMM 2 facility baseline emissions beyond 2025, the facility's 2030 requirement will be increased by an amount depending on the amount of time the facility emits beyond the GEMM 2 GHG baseline emissions. This ensures any excess emissions past 2025 are mitigated and displaced in 2030 and beyond, eventually achieving a greater cumulative reduction over time.

Pursuant to Part B, Section III.B, covered GEMM 2 facilities that are unable to achieve the required GHG reductions through onsite measures can comply by retiring GHG credits. GEMM 2 facilities may generate GHG credits by reducing GHG emissions below a facility's 2030 GHG reduction requirement including, beginning in 2031, through direct air capture projects, or may purchase GHG credits from other facilities.

Part B, Section I.B of the rule applies separate requirements to existing facilities that report direct GHG emissions of equal to or greater than 25,000 metric tons of CO₂e after the effective date of the rule. This includes any EITE subject to the rule that exceeds the lowered 25,000 metric tons of CO₂e threshold which elects to comply with Regulation Number 27 through Part B, rather than through Part C. As of the effective date of the rule, there was one EITE facility in Colorado, Golden Aluminum Inc., in this situation. Golden Aluminum will be subject to the requirements established in Part B, Section I.A., unless it elects to comply with Regulation Number 27 through Part C. The historical emissions of this facility are known and were included in the calculations for the covered GEMM 2 facilities in the development of the GEMM 2 GHG emission reduction requirements. The GEMM 2 regulation has been drafted to assure covered GEMM 2 facilities achieve a 20% reduction in GHG emissions compared to 2015 whether Golden Aluminum chooses to comply with Regulation Number 27 through Part B or Part C.

Carbon Capture and Storage

Under Part B, Section III.A.1, a GEMM 2 facility may account for direct emission reductions from a carbon capture and storage system (CCS), with capture of CO₂ performed onsite at the GEMM 2 facility, for purposes of compliance with the facility's GEMM 2 annual GHG reduction requirement in any year. However, prior to using reductions from such systems towards a facility's compliance, the Division must approve a protocol applicable to such systems. The Division assessed three published protocols and one draft protocol, along with two sets of accounting methodologies, related to CCS.

The three published protocols were issued by the American Carbon Registry (see American Carbon Registry, Carbon Capture and Storage Projects, <https://americancarbonregistry.org/carbon-accounting/standards-methodologies/carbon-capture-and-storage-in-oil-and-gas-reservoirs>), the California Air Resource Board (see California Air Resources Board, Carbon Sequestration: Carbon Capture, Removal, Utilization, and Storage, <https://ww2.arb.ca.gov/our-work/programs/carbon-sequestration-carbon-capture-removal-utilization-and-storage>), and the Government of Alberta, Canada (see Alberta Government, Quantification protocol for CO₂ capture and permanent storage in deep saline aquifers, <https://open.alberta.ca/publications/9780778572213>).

The one draft protocol was issued by the Verified Carbon Standard (see Verified Carbon Standard, Draft Methodology for Carbon Capture and Storage, <https://verra.org/wp-content/uploads/2023/06/CCS-Methodology-Public-Consultation-Draft.pdf>).

The two accounting methodologies were the 2006 IPCC Guidelines (see 2006 IPC Guidelines, https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_5_Ch5_CCS.pdf) and 40 CFR, Part 98 (Part 98). Additionally, the Division assessed the 2005 IPCC Special Report on CCS (see IPCC, Carbon Dioxide Capture and Storage, <https://www.ipcc.ch/report/carbon-dioxide-capture-and-storage/>).

Although certain of these protocols and methodologies showed promise for use in the context of the Commission's GHG reduction rules, as of the date of rule adoption, there was no existing published protocol that could be adopted by reference, and no single accounting methodology that could be referenced, for such use.

As such, the Commission directs the Division to establish or adopt by reference a standardized CCS protocol or protocols in consultation with interested stakeholders and consistent with the CCUS Task Force Subcommittee Recommendations. Any such protocol(s) must include, without limitation, the following requirements:

1. CO₂ must be captured onsite at a GEMM 2 facility. The protocol should reflect that as long as the carbon capture is taking place onsite, the storage/sequestration need not also take place onsite.
2. CO₂ captured need not be from a combustion or process emission source category as defined by 40 CFR, Part 98.
3. Sequestered Biogenic CO₂ shall be counted as a negative emission.
4. Only permanent geologic sequestration in a UIC Class VI well, or other forms of permanent sequestration is permitted.
5. CO₂ captured by a GEMM 2 facility must either remain in the custody of the facility or custody transfers must be detailed to an extent permitting the complete traceability from the point of capture to the point of permanent sequestration as set forth in the protocol.
6. The GEMM 2 facility shall be responsible for compiling and reporting all onsite emissions as well as all direct Scope 1 emissions attributable to the capture, conditioning, transport, and storage of CO₂ regardless of whether those emissions were incurred by any downstream entity(ies).
7. Any GHG credits issued for CCS will be equal to the total annual metric tons of CO₂ sequestered in subsurface geologic formations at the facility in the reporting year as calculated using Equation RR-12 of Subpart RR of 40 CFR, Part 98, or comparable methodology, minus the total mass of emissions, in CO₂ equivalents, incurred in the capture, conditioning, and transport of the CO₂ from the capture facility through to the receiving flow meter at the sequestration well or site that would not have occurred in the absence of the project. Emission sources and respective accounting approaches that must be accounted for include, without limitation:
 - 7.1. Greenhouse gas emissions resulting from the combustion of fossil fuels for the powering of stationary equipment used in the capture, and transport via pipeline of CO₂, shall be reported per Subpart C of 40 CFR, Part 98, or per a comparable methodology.
 - 7.2 Excluding the CO₂ captured or transported for the purpose of sequestration, greenhouse gas emissions resulting from leaks, vents and flares from stationary equipment of GHGs other than the CO₂ captured or transported for the purpose of sequestration shall be reported per Subpart W of 40 CFR, Part 98, or per a comparable methodology.
 - 7.3 Greenhouse gas emissions resulting from vehicle transportation for the delivery of CO₂ via containers from a GEMM 2 capture facility to a well shall be accounted for by tracking fuel use attributed to that transportation for both pick-up and delivery of the CO₂ and applying the appropriate emission factor for the fuel type as found in Table MM-1 or Table MM-2 of Subpart MM of 40 CFR, Part 98, or a comparable appropriate emission factor.

State-Managed GHG Reduction Fund

The Commission directs the Division to propose establishment of a state-managed fund to receive and allocate monies to finance projects to reduce GHG emissions from the industrial and manufacturing sector by no later than September 2025, and to ask the Commission to hold a rulemaking hearing to establish this fund, the hearing to be held no later than December 2025. This state-managed fund could serve as a compliance option for any GEMM 2 facility unable to comply by other means laid out in the rule, allowing the facility to instead pay fees to this fund on a per metric ton of CO₂e basis up to the amount required to achieve the facility's reduction requirement for that year. The Commission expects that fees would need to be set above any regulatory price cap for onsite reductions. The purpose of the fund should be to finance projects at other industrial or manufacturing sites located within Disproportionately Impacted Communities, or finance otherwise cost-prohibitive onsite reduction projects within the group of regulated entities in the industrial and manufacturing sector. The reduction projects funded should be prioritized by considering both the GHG reductions and co-pollutant reductions estimated from each project, and prioritized within communities surrounding the GEMM 2 facilities that have the highest EnviroScreen scores to ensure the projects are occurring in the most vulnerable communities.

GHG Reduction Plan

As established in Part B, Section II.A., by September 30, 2025, each covered GEMM 2 facility (except for glass manufacturing facilities) must develop a GHG reduction plan showing how the facility will achieve the required GHG reductions and how co-pollutant emission reductions were considered and prioritized in accordance with the rule. For glass manufacturing facilities, the GHG reduction plans are due by June 1, 2027. The facility will be required to implement the portfolio of onsite measures with an average cost equal to or below the 2030 social cost of GHGs (as established in the Federal Interagency Working Group's February 2021 report on the Social Cost of Greenhouse Gases), which achieves the greatest reduction in harmful air pollution, up to the facility's 2030 GHG emission requirement before it may use the credit system to comply with the rule. The average cost is determined by dividing total cost of the portfolio by the total emission reduction. The 2030 social cost of GHG values are, for carbon dioxide, \$89 per metric ton of carbon dioxide; for methane, \$2,500 per metric ton of methane; and for nitrous oxide, \$33,000 per metric ton of nitrous oxide, with other GHGs converted to CO₂e. To determine whether a portfolio of measures exceeds this cost, facilities are required to show that the cost of implementing the portfolio of measures, over the lifetime of the equipment involved, will have an average cost of more than the 2030 social cost of the respective GHG gases abated.

The Commission determined that the 2030 social cost of GHGs is an appropriate regulatory price cap for requiring onsite measures through a portfolio approach. The social cost of GHGs, however, is not intended to be a price above which the Commission considers GHG emission reduction measures to be cost-prohibitive nor as a ceiling above which facilities should not implement GHG emission reduction measures that are cost-effective for those facilities, as cost-effectiveness varies for each facility. A portfolio of measures may require inclusion of individual measures above this cost, where the average cost of the portfolio remains equal to or less than the 2030 social cost of GHGs. The 2030 social cost of GHGs was considered an appropriate average cost because the social cost of GHGs, as defined in the Act at § 25-7-110.5(4)(f), C.R.S., is a comparative tool that is already established and defined as cost-effective since it is the cost at which the net benefits of GHG emission reduction measures generally outweigh the costs. Using the 2030 social cost of GHGs value additionally aligns with the year the reductions are required to be achieved by the covered GEMM 2 facilities, and the year around which the GHG reduction plan is structured. Above this portfolio cost, the Commission encourages facilities to consider the cost-effectiveness of GHG emissions reductions measures for their individual facilities by requiring facilities to identify and price all technically feasible GHG emission reduction measures and establishing a credit trading program, both which help to create the business case to implement measures above the 2030 social cost of GHGs if they are cost-effective for facilities independently. Requiring facilities to assess the cost of all measures also allows for measures to be assessed for accuracy.

This requirement also allows the Division to collect information on the types and cost of reduction measures that could potentially become cost-effective in the future, or with additional financing through incentives and grant programs and for the independent third party to verify that they are, in fact, accurate. The Commission's approach described above will help to prevent GHG emissions leakage out of state, which is a concern if a facility would otherwise be required to decrease production to comply with Regulation Number 27. The Commission is aware that the social cost of GHG values might be updated upwards by the Federal Interagency Working Group. The Commission asks the Division to monitor any such updates and return to the Commission with a proposal to update the SC-GHG used in the rule for purposes of cost-effectiveness if the SC-GHG is increased by the Federal Interagency Working Group.

In addition to prioritizing onsite GHG reductions, which is expected to result in reductions of harmful air pollution, the rule satisfies the statutory requirement in § 25-7-105(1)(e)(XIII), C.R.S. to include protections for disproportionately impacted communities and prioritize emission reductions that reduce harmful air pollution that adversely affects disproportionately impacted communities through two additional mechanisms set out in Part B, Section II.A.3.a. and Part B, Section II.A.6. In Part B, Section II.A.3.a., the Commission required that facilities must propose to implement the GHG reduction measures with the greater reduction in harmful air pollution for measures that yield GHG emission reductions within 5% of each other and are at or below the 2030 social cost of GHGs. This prioritizes GHG reduction measures that also reduce the greater amount of harmful air pollution for the benefit and protection of disproportionately impacted communities. In Part B, Section II.A.6, the Commission adopted a mechanism by which facilities that use the GHG credit trading system to reach their 2030 GHG emission requirements will be required to "true-up" their onsite reductions of harmful air pollution.

For this process, the Commission adopted a higher regulatory price cap for implementing measures to reduce harmful air pollution because, upon proposing to use the GHG credit trading system, a covered GEMM 2 facility will have already proposed the portfolio of measures up to the 2030 social cost of GHGs and concluded that no additional measures are available under that cost. Therefore, by increasing the regulatory price cap threshold to 50% above the 2030 social cost of GHGs, additional GHG reduction measures that also reduce harmful air pollution that are not included in the facility's portfolio of measures may be identified and used to quantify corresponding harmful air pollution reductions. The calculation of the portfolio cost does not include additional measures identified and included under Part B, Section II.A.6. These requirements were structured to directly address the statutory requirements to provide protections for disproportionately impacted communities and prioritize reductions of harmful air pollution. To allow for a consistent methodology to be used in quantifying and comparing co-pollutant reductions, the Commission directs the Division to publish a guidance document outlining the specific process that should be followed. The guidance document will aid regulated sources as well as third-party reviewers in ensuring a consistent method is used in the development of the GHG reduction plans.

Under Part B, Sections II.C and II.D, an independent third party must review and certify the facility's GHG reduction plan. Owners and operators of GEMM 2 facilities will pay the full cost of the independent third-party review and certification for their facility. In contrast to the qualified third-party auditor requirements under Part C, Section I.A.3., the independent third party will be contracted by the State of Colorado to assure objectivity and neutrality of the third-party review and certification. Under Part B, Section IV, covered GEMM 2 facilities must also submit annual emission reports by March 31 of each calendar year starting in 2025, and compliance certifications by September 30 for each compliance period starting in 2027. The facility must certify a facility's annual emissions, provide updates on reducing air pollution where located near or in a disproportionately impacted community, and document non-facility achieved GHG reductions (e.g., purchased GHG credits).

When calculating its emissions reduction targets and submitting its annual compliance certification, a GEMM 2 facility may account for a percent of the emissions avoided by utilization of a combined heat and power ("CHP") unit, or cogeneration unit. The Commission recognizes that industrial CHP can provide significant GHG emissions reductions in the near- to mid-term as marginal grid emissions continue to be based on a mix of fossil fuels in Colorado.

Additionally, cogeneration units avoid losses associated with conventional electricity supply, which further reduces fuel use, helps avoid the need for new transmission and distribution infrastructure, and eases grid congestion when demand for electricity is high, particularly here in Colorado. The Commission also notes that in the long-term, consistent with a U.S. Department of Energy “Industrial Decarbonization Roadmap” (Sept. 2022), CHP systems can be retrofitted to use clean fuels, and so it makes little sense to incentivize early shutdown of such systems currently. Therefore, the Commission adopted a provision allowing a GEMM 2 facility with a combined heat and power unit to reduce its GEMM 2 annual GHG emissions requirement in 2024 through 2029, based on the amount of displaced on-site emissions associated with boiler usage, up to 50% of the facility’s annual GHG emissions reduction requirement. The Commission intends that credit given for CHP sunset as of December 31, 2029. The Commission intends that the combined heat and power compliance credit will be available only so long as a facility demonstrates its displaced on-site thermal energy emissions based on the six-step formula (below) contained in the Division-approved form that calculates displaced emissions. This ensures that the State achieves both the GEMM 2 specific target and the State’s economy-wide target in a more efficient manner by avoiding simply shifting GHG emissions to another industrial sector.

Step 1:	Displaced electricity emissions	D_E	$D_E = (C_P + G_L) \times G_E$ where: C_P = CHP electricity production G_L = Electric grid transmission & distribution loss G_E = Electric grid emission factor
Step 2:	Displaced thermal emissions	D_T	$D_T = C_T / T_P \times T_E$ where: C_T = Utilized CHP thermal output T_P = Displaced thermal production efficiency T_E = Displaced thermal emission factor
Step 3:	Displaced utility emissions	D_U	$D_U = D_E + D_T$ where: D_E = Displaced electricity emissions D_T = Displaced thermal emissions
Step 4:	CHP emissions	C_E	$C_E = C_F + F_E$ where: C_F = CHP fuel consumption F_E = CHP fuel emission factor
Step 5:	Total avoided emissions	A_T	$A_T = D_U - C_E$ where: D_U = Displaced utility emissions C_E = CHP emissions
Step 6:	Direct stationary avoided emissions	A_D	$A_D = D_T / D_U \times A_T$ where: D_T = Displaced thermal emissions D_U = Displaced utility emissions A_T = Total avoided emissions

Transparency

In line with the Commission's commitment to equitable representation and meaningful community engagement, the Commission directs the Division to take the following actions related to transparency in its implementation of Regulation Number 27.

The Division is directed to conduct the public meetings required under Part B, Section II.I. consistent with the spirit of the outreach requirements provided in the Environmental Justice Act including (1) holding multiple meetings at variable times such as weekend, evening, and/or morning sessions; (2) providing at least 30 day public notice before any public input opportunity; (3) disseminating meeting announcements through different methods and community organizations; and (4) hosting meetings in multiple locations, such as urban centers, rural locations. The Division is further directed to gather public input on GHG reduction plans via a variety of methods, including in person, virtual/online, online comment portal/email, and call-in. Colorado residents participating in such public meetings outside of paid employment will be offered participation stipends as an expression of gratitude for their time and child care stipends to promote accessible, meaningful involvement in the process. Such public meetings will be announced in English and Spanish and interpretation will be provided in languages other than English upon request.

Additionally, the Division is directed to post to its website a plain-language description of the contents of the following documents in the top two languages spoken by the communities surrounding each of the GEMM 2 facilities: (1) GHG reduction plans (Part B, Section II.A); (2) annual compliance certifications (Part B, Section IV.A); (3) certifications that a facility will comply with its 2030 requirement in 2025 (Part B, Section I.A.6); (4) records related to compliance with a facility's approved GHG reduction plan (Part B, Section V.A.2); and (5) credit trading program registration applications (Part D, Section II.A.1). The Division will provide translated copies of these documents to members of the public upon request or will provide a meeting with a Division subject-matter expert to review requested document(s) with an interpreter present.

Recordkeeping

Certain submissions required of GEMM 2 facilities may contain confidential business information (CBI). Part B, Section V includes a requirement that any potential CBI submitted by GEMM 2 facilities must be clearly identified and be submitted in a separate, supplementary document. The Commission does not, however, intend that such statements be determinative of whether the information is in fact CBI under Colorado law, but expects that information will be made available as required and permitted by the Colorado Open Records Act.

Greenhouse Gas Credit Trading System

The Commission also expanded upon the GHG accounting and tracking system, as required in § 25-7-105(f)(II), C.R.S., with the GHG credit trading system set out in Part D.

The GHG credit trading system allows "regulated sources" to generate and trade or retire GHG credits as a compliance mechanism for this rule. As "regulated sources," an EITE facility may generate one GHG credit per metric ton of CO₂e reduced beyond its annual emissions limitation in the relevant year which it may trade with other EITE sources. A GEMM 2 facility may generate one GHG credit per metric ton of CO₂e reduced beyond its 2030 GHG emissions requirement in the given year which it may trade with other GEMM 2 facilities. Before allowing trading between EITE and GEMM 2 sources, the Commission directs the Division to engage in a stakeholder process or technical working group to publish guidance by December 1, 2024. Such guidance must ensure that trading between EITE and GEMM 2 sources does not compromise the sector's achievement of the GHG emissions reduction requirement in § 25-7-105(1)(e)(XIII), C.R.S. As with any credit trading market, it is important that this GHG credit trading system ensure that one GHG credit equates to one metric ton of CO₂e reduced to ensure the GHG credits in the market accurately reflect GHG emission reductions.

Because EITE facilities generate GHG credits on an intensity basis, one GHG credit generated by an EITE facility does not always equate to one metric ton of CO₂e reduced such that trading between EITE and GEMM 2 generated GHG credits may jeopardize the State's progress towards its climate goals. Therefore, until a framework could be established to prevent jeopardizing the State's progress towards its climate goals, GHG credit trading between EITE and GEMM 2 facilities was restricted until 2025 after the Division published guidance to ensure the sector aligns with its climate targets in § 25-7-105, C.R.S. By ensuring that one metric ton of CO₂e reduced is equal to one GHG credit, which a facility can bank for up to three years and can retire or trade as needed, the GHG credit trading system established in Part D was structured to (1) not jeopardize the State's progress towards its climate goals, and (2) allows a facility to retain the full value of one metric ton of CO₂e reduced at the facility in the GHG credit trading system.

The GHG credit program also incentivizes near-term emission reductions, as required by the Act, because it allows a facility to retain the full value of one metric ton of CO₂e reduced at the facility in the credit market while encouraging facilities to avoid additional costs to purchase GHG credits and creating a profitable business opportunity for facilities that can reduce their emissions beyond their allowances.

In addition, beginning in 2031, GEMM 2 facilities may generate one GHG credit per metric ton of CO₂e quantifiably reduced through offsite direct air carbon capture projects. The Commission directs the Division to approve a protocol governing the implementation of such projects. This provision was intended to encourage direct air capture projects that remove carbon from the air, which will help Colorado achieve its climate change mitigation goals. See § 25-7-102(2), C.R.S. Any GHG credit generated through a direct air carbon capture project will be scrutinized to ensure that GHG emissions reductions are real, quantifiable, permanent, verifiable, enforceable, and that the ratio of one GHG credit to one metric ton of CO₂e reduction is maintained. Offering an additional alternative compliance option for GEMM 2 facilities aligns with the Commission's ability to include regulatory strategies "that enhance cost-effectiveness" and "compliance flexibility" in its rules per § 25-7-105(1)(e)(V), C.R.S. Importantly, this option does not undermine the rule's requirement for the sector to reduce GHG emissions by 20% from 2015 levels because it is available for compliance after facilities are required to initially achieve this level of GHG emission reductions. Nor does it undermine protections for disproportionately impacted communities because it is subject to the rule's requirements that, (1) under Part B, Section II.A.4, facilities implement all onsite reductions up to the 2030 social cost of GHGs before using the GHG credit trading system towards achieving their 2030 reduction requirements, and that (2) facilities using the GHG credit trading system that are located near residential disproportionately impacted communities reduce harmful air pollution under Part B, Section II.A.5.

The GHG credit trading system will be operational and open to EITE and GEMM 2 facilities for GHG credit trading by December 1, 2024. The Commission anticipates that this will be sufficient time for the Division to develop, test, and launch the GHG credit trading system for facilities to utilize GHG credits as a form of compliance beginning in 2025 for the 2024 compliance year. The Division will issue GHG credits annually to align with the annual emissions compliance obligations. Each GHG credit will be uniquely identifiable to assist the Division with tracking GHG credit movement within the system and preventing double-counting of GHG emission reductions. Also to prevent double-counting of GHG emission reductions, the GHG credit trading system will prohibit the transfer or use of GHG credits generated within the system to other external credit trading systems to assure the GHG credits are used and accounted for within Colorado. Finally, unless retired earlier for compliance purposes, all GHG credits will expire three years from the date they are generated. Allowing facilities to hold GHG credits for three years will allow facilities to time the sale of their GHG credits with the market and to generate revenue over time by selling a greater number of GHG credits after they have had an opportunity to bank GHG credits. Three years is the lifetime of a GHG credit to create a balance of supply and demand for GHG credits in the system and was based on the Division's projections of GHG credit availability in the GHG credit market each year.

The Commission heard testimony during the hearing from individuals, community groups and officials expressing concern that the regulations as proposed did not go far enough to provide the necessary reductions of co-pollutants in Disproportionately Impacted Communities, as well as testimony expressing the contrary concern that the regulations could adversely affect the production output and number of jobs in those industries covered by the rule. The Commission appreciates hearing from all who provided comments regarding the potential impacts and issues associated with this rulemaking. This rulemaking was directed by the legislature both to ensure greenhouse gas emission reductions from the industrial and manufacturing sector and to prioritize emission reductions that will reduce emissions of co-pollutants that adversely affect Disproportionately Impacted Communities. In adopting this rule, the Commission has worked to balance the conflicting concerns raised in the testimony. Recognizing that the rule presents new concepts and requirements that may need to be reevaluated to ensure the goals of the legislation and the rule's programs are met, the Commission:

1. Directs the Division to evaluate the trading program and to identify any modifications that may be necessary. If any such modifications are identified by the Division, the Commission requests that the Division bring a petition to request a rulemaking hearing to the Commission no later than September 2025.
2. Directs the Division to report to the Commission on or before December 31, 2025 on:
 - a. The status of the trading program;
 - b. The co-pollutant reductions associated with the credits generated for the trades;
 - c. Updated projections for the 2026 credit market, and likely compliance pathways for the GEMM 2 facilities.
 - d. The 2022- 2024 GHG emissions from both the GEMM 1 and GEMM 2 facilities
 - e. The GHG plans submitted by the covered facilities and information on emissions reduction strategies in the plans including co-pollutant reductions.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Additional Considerations

The following are additional findings of the Commission made in accordance with the Act:

§ 25-7-110.5(5)(b), C.R.S.

As these revisions exceed and may differ from the federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), C.R.S., the Commission determines:

- (I) Any federal requirements that are applicable to this situation with a commentary on those requirements;

Manufacturing sector stationary sources are required to report GHG emissions under existing federal regulations. The Mandatory Reporting Rule requires sources with annual emissions equal to or greater than 25,000 metric tons of CO₂e per year to report through the EPA's Greenhouse Gas Reporting Program. Some specific source types are considered "all in" and required to report GHG emissions even if they are under the 25,000 metric ton per year threshold. However, emission reduction requirements such as those in Regulation Number 27 are not part of any federal law or regulation.

- (II) Whether the applicable federal requirements are performance-based or technology-based and whether there is any flexibility in those requirements, and if not, why not;

The applicable federal requirements are not performance-based or technology-based because they are reporting requirements only. Manufacturing sector stationary sources are required to report GHG emissions under existing federal regulations. The Mandatory Reporting Rule requires sources with annual emissions equal to or greater than 25,000 metric tons of CO₂e per year to report through the EPA's Greenhouse Gas Reporting Program. Some specific source types are considered "all in" and required to report GHG emissions even if they are under the 25,000 metric ton per year threshold.

- (III) Whether the applicable federal requirements specifically address the issues that are of concern to Colorado and whether data or information that would reasonably reflect Colorado's concern and situation was considered in the federal process that established the federal requirements;

There are no federal requirements that specifically address the issues that are of concern to Colorado.

- (IV) Whether the proposed requirement will improve the ability of the regulated community to comply in a more cost-effective way by clarifying confusing or potentially conflicting requirements (within or cross-media), increasing certainty, or preventing or reducing the need for costly retrofit to meet more stringent requirements later;

The adopted revisions give meaningful effect to § 25-7-105(1)(e)(XIII), C.R.S., and provide regulated entities flexibility to identify and cost-effectively employ emissions control technologies with guidance of a regulatory price cap. The covered GEMM 2 facilities are allowed to comply using the GHG credit trading system to allow for flexibility.

- (V) Whether there is a timing issue which might justify changing the time frame for implementation of federal requirements;

There are no applicable federal requirements that create any timing issues. Part B allows regulated entities a reasonable time to comply with the reduction requirements and to submit the GHG reduction plan. It also allows opportunities for GEMM 2 facilities to use the GHG credit trading system opportunities for alternative compliance.

- (VI) Whether the proposed requirement will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth;

Regulation Number 27 does not set constraints on production for covered facilities. Regulated entities may use the GHG credit trading system after onsite measures are employed. Regulated entities may also use the program if the facility must increase in direct GHG emissions as a result of increased production. In addition, regulated entities may pay into a state-managed GHG reduction fund for purposes of compliance, if and when such a fund is established.

- (VII) Whether the proposed requirement establishes or maintains reasonable equity in the requirements for various sources;

All covered GEMM 2 facilities are subject to the GHG reduction requirements depending on multiple, equity-focused factors. The proposal assigns tiered reduction requirements based largely on the facilities' achieved reductions in an equitable manner. Those that had achieved significant GHG emission reductions since 2015 have a lesser 2030 reduction obligation than those that had reduced less than 20%, or increased emissions. Additionally, facilities that are the larger emitters relative to the group's total emissions have an additional percentage of reduction required because of the impact the facilities have on the group reaching a 20% reduction from 2015 by 2030, and facilities with more emissions tend to have more opportunities to reduce GHG emissions, versus the smaller facilities in the group.

- (VIII) Whether others would face increased costs if a more stringent rule is not enacted;

The General Assembly has acknowledged that climate change impacts Colorado's economy and directed that GHG emissions should be reduced across all sectors of our economy. Colorado has established specific GHG reduction goals. Reductions not achieved in one sector will require measures in other sectors of the economy to achieve the state's GHG reduction goals.

Furthermore, the General Assembly provided requirements that the industrial sector in Colorado reduce its GHG emissions by 20% by 2030 compared to what it emitted in 2015. The GEMM 2 rule is addressing emissions from the manufacturing sector, which account for approximately one-third of the industrial sector's total emissions. The facilities regulated under both GEMM 1 and the proposed revisions of GEMM 2 account for 75% of the manufacturing sector's emissions. Reductions not timely realized by these facilities will require additional measures to achieve reductions in other industrial sources, many of which are already regulated for GHG emissions, to reach the state's industrial sector 2030 target.

- (IX) Whether the proposed requirement includes procedural, reporting, or monitoring requirements that are different from applicable federal requirements and, if so, why and what the "compelling reason" is for different procedural, reporting, or monitoring requirements;

Regulation Number 27 gives effect to the General Assembly's adoption of § 25-7-105(1)(e)(XIII), C.R.S. which is a unique requirement of Colorado law. The "compelling reason" for the GHG reduction plans and annual compliance reports required under Part B of Regulation Number 27 is to ensure satisfaction of § 25-7-105(1)(e)(XIII), C.R.S. Such plans and reports will also inform the state's strategies and future regulations to accomplish the statewide GHG pollution reduction goals and address the impacts of climate change set forth in § 25-7-102(2), C.R.S. and further sector-specific emission reductions under § 25-7-105(1)(e)(XIII), C.R.S.

- (X) Whether demonstrated technology is available to comply with the proposed requirement;

Regulation Number 27 does not require the use of any specific technology but instead serves as a mechanism to assure reductions are achieved by specific manufacturing sources by setting individual GHG reduction requirements for applicable facilities and allowing a GHG credit trading system to reach those targets. Regulation Number 27 also does not require the use of any specific technology for reducing harmful air pollution, but instead prompts regulated entities to evaluate the control technologies that may have such results. The GHG reduction plans are used to conduct this evaluation and must include analyses of, but not necessarily implementation of, transformative technologies. All measures identified in the GHG reduction plans will be based on demonstrated and available technologies.

- (XI) Whether the proposed requirement will contribute to the prevention of pollution or address a potential problem and represent a more cost-effective environmental gain;

The revisions to Regulation Number 27 enable the Commission to require specific mass-based reduction requirements for large manufacturing facilities. The covered GEMM 2 facilities are required to comply first with technically feasible onsite reduction measures up to the 2030 social cost of GHGs, and if they are not able to achieve their reduction goal utilizing onsite measures, they may use the GHG credit trading system.

The GHG emissions reductions from this rule are expected to help Colorado achieve the statewide GHG pollution reduction goals in § 25-7-102(2)(g), C.R.S., and the sector-specific GHG emission reductions set forth in § 25-7-105(1)(e)(XIII), C.R.S. Anticipated reductions in co-pollutants are expected to have positive health benefits for the people of Colorado.

- (XII) Whether an alternative rule, including a no-action alternative, would address the required standard.

This rule implements the statutory requirements of §§ 25-7-105(1)(e)(XIII), C.R.S. Alternatives exist for how to accomplish these requirements, including different emission thresholds for qualifying entities, different standards for evaluating GHG reduction plans and measures, and the provision of no or differing means of alternative compliance as well as different timing requirements for emission reductions. The Commission determined that Part B appropriately gives effect to the statutory requirements and is consistent with the statewide and sector-specific GHG pollution reduction goals. A no-action alternative is not available under § 25-7-105(1)(e), C.R.S.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

- (I) These rules are based on reasonably available, validated, reviewed, and sound scientific methodologies and all validated, reviewed, and sound scientific methodologies and information made available by interested parties has been considered.
- (II) Evidence in the record supports the finding that the rule shall result in a demonstrable reduction in GHG pollution and co-pollutants and will enable the Commission to satisfy the requirements of §§ 25-7-102, -105(1)(e), -106, and/or -109, C.R.S., as applicable.
- (III) Evidence in the record supports the finding that the rule shall bring about reductions in risks to human health and the environment that will justify the costs to government, the regulated community, and to the public to implement and comply with the rule.
- (IV) The rules are the most cost-effective to achieve the necessary and desired results and reduction in air pollution.
- (V) The rule will maximize the air quality benefits of regulation in the most cost-effective manner.

V. Adopted: December 18-20, 2024

Expansion of Part D, Greenhouse Gas Credit Trading, to recovered methane and the midstream oil and gas segment.

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-103(4), C.R.S., the Colorado Air Pollution Prevention and Control Act, §§ 25-7-110 and -110.5, C.R.S., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

In 2021, the Commission adopted requirements for upstream and midstream segment oil and gas operations, to reduce greenhouse gas (GHG) emissions from those operations to achieve the GHG reduction requirements of House Bill 21-1266. Specifically for the midstream segment, the Commission adopted requirements for owners or operators to submit fuel combustion equipment company emission reduction plans to a midstream steering committee by July 31, 2022; for the midstream steering committee to develop recommendations for a midstream segment emission reduction plan (sERP) and submit the recommendations to the Division by March 31, 2024; and for the Division to submit a regulatory proposal to the Commission by August 31, 2024, and request a rulemaking hearing for no later than December 31, 2024. The Commission adopted in Regulation Number 7, Part B, Section VII. a midstream segment emission reduction program that will achieve a 20% reduction in CO₂e from the 2015 baseline, as required by § 25-7-140(2)(a)(II). As part of this program, the Commission also adopted provisions for GHG crediting and trading. Therefore, the Commission adopted revisions to Regulation Number 27 expanding Part D to midstream companies.

Specific Statutory Authority

The Colorado Air Pollution Prevention and Control Act (Act), specifically § 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102, C.R.S., and that are necessary for the proper implementation and administration of the Act. § 25-7-105(1)(e), C.R.S., authorizes the Commission to promulgate implementing rules and regulations consistent with the statewide GHG pollution reduction goals in § 25-7-102(2)(g), C.R.S. § 25-7-106, C.R.S., provides the Commission “maximum flexibility in developing an effective air quality program and [promulgating] such [a] combination of regulations as may be necessary or desirable to carry out that program.” § 25-7-106(6), C.R.S., further authorizes the Commission to require owners and operators of any air pollution source to monitor, record, and report information.

Purpose

The Commission expanded the Part D Greenhouse Gas Credit Trading System to both midstream companies and recovered methane. Concerning midstream companies, the Commission adopted specific applicability, timing, and other requirements in Regulation Number 7, Part B, Section VII. Concerning recovered methane, the addition of registration requirements for entities generating and obtaining recovered methane credits in Part D, Sections II.A. and II.C. are meant to align with existing requirements in Regulation 22, Part C, Section I.D.2.a.(ii), which references Part D, Section II.C.

The Commission also updated the system terminology from “credit trading system” to “crediting and tracking system” to be more accurate with the system’s purpose and intent. Corresponding terminology updates were included in Regulation Number 7, Part B, Section VII. for the midstream segment emission reduction program. The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Further, these revisions will include any typographical, grammatical and formatting errors throughout the regulation.

Incorporation by Reference

Section 24-4-103(12.5) of the State Administrative Procedure Act allows the Commission to incorporate by reference federal regulations. The criteria of §24-4-103(12.5) are met by including specific information and making the regulations available because repeating the full text of each of the federal regulations incorporated would be unduly cumbersome and inexpedient. To fully comply with these criteria, the Commission includes, where necessary, reference dates to rules and reference methods incorporated in Regulation Number 27.

Additional Considerations

The following are additional findings of the Commission made in accordance with the Act: § 25-7-110.5(5)(b), C.R.S.

As these revisions exceed and may differ from the federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), C.R.S., the Commission determines:

- (I) Any federal requirements that are applicable to this situation with a commentary on those requirements;

Stationary sources are required to report GHG emissions under existing federal regulations. The Mandatory Reporting Rule requires sources with annual emissions equal to or greater than 25,000 metric tons of CO₂e per year to report through the EPA's Greenhouse Gas Reporting Program. However, there is currently no federal level greenhouse gas crediting and tracking system.

- (II) Whether the applicable federal requirements are performance-based or technology-based and whether there is any flexibility in those requirements, and if not, why not;

The applicable federal requirements are not performance-based or technology-based because they are reporting requirements only.

- (III) Whether the applicable federal requirements specifically address the issues that are of concern to Colorado and whether data or information that would reasonably reflect Colorado's concern and situation was considered in the federal process that established the federal requirements;

There are no federal requirements that specifically address the issues that are of concern to Colorado.

- (IV) Whether the proposed requirement will improve the ability of the regulated community to comply in a more cost-effective way by clarifying confusing or potentially conflicting requirements (within or cross-media), increasing certainty, or preventing or reducing the need for costly retrofit to meet more stringent requirements later;

The adopted revisions give meaningful effect to § 25-7-105(1)(e)(XIII), C.R.S., and provide regulated entities flexibility to identify and cost-effectively employ emissions control technologies. The covered midstream companies are allowed to comply using the GHG crediting and tracking system to allow for flexibility.

- (V) Whether there is a timing issue which might justify changing the time frame for implementation of federal requirements;

There are no applicable federal requirements that create any timing issues.

- (VI) Whether the proposed requirement will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth;

Regulation Number 27 does not set constraints on production for covered facilities. Regulated entities may use the GHG credit trading system after onsite measures are employed. Regulated entities may also use the program if the facility must increase in direct GHG emissions as a result of increased production.

- (VII) Whether the proposed requirement establishes or maintains reasonable equity in the requirements for various sources;

The rule maintains reasonable equity for midstream companies as crediting and tracking requirements and opportunities are the same for each company.

- (VIII) Whether others would face increased costs if a more stringent rule is not enacted;

The General Assembly has acknowledged that climate change impacts Colorado's economy and directed that GHG emissions should be reduced across all sectors of our economy. Colorado has established specific GHG reduction goals. Reductions not achieved in one sector will require measures in other sectors of the economy to achieve the state's GHG reduction goals.

- (IX) Whether the proposed requirement includes procedural, reporting, or monitoring requirements that are different from applicable federal requirements and, if so, why and what the "compelling reason" is for different procedural, reporting, or monitoring requirements;

There are no comparable federal GHG crediting and tracking system requirements.

- (X) Whether demonstrated technology is available to comply with the proposed requirement;

Regulation Number 27 does not require the use of any specific technology but instead serves as a mechanism to allow GHG crediting and trading to reach applicable emission reduction targets.

- (XI) Whether the proposed requirement will contribute to the prevention of pollution or address a potential problem and represent a more cost-effective environmental gain;

The revisions to Regulation Number 27 enable a midstream company to use the GHG crediting and tracking system if they are not able to achieve their reduction goal utilizing onsite measures.

The GHG emissions reductions from this rule are expected to help Colorado achieve the statewide GHG pollution reduction goals in § 25-7-102(2)(g), C.R.S., and the sector-specific GHG emission reductions set forth in § 25-7-105(1)(e)(XIII), C.R.S. Anticipated reductions in co-pollutants are expected to have positive health benefits for the people of Colorado.

- (XII) Whether an alternative rule, including a no-action alternative, would address the required standard.

The expansion of the GHG crediting and tracking system to the midstream segment is needed to achieve the statutorily mandated emission reductions. As noted, the State Air Act requires the Commission to implement GHG emission reduction strategies in order to secure reductions of pollution consistent with the statewide GHG emission reduction goals. Currently emissions projections over the next decade demonstrate that a no-action alternative would fall short of achieving Colorado's reduction goals.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

- (I) These rules are based on reasonably available, validated, reviewed, and sound scientific methodologies and all validated, reviewed, and sound scientific methodologies and information made available by interested parties has been considered.
- (II) Evidence in the record supports the finding that the rule shall result in a demonstrable reduction in GHG pollution and co-pollutants and will enable the Commission to satisfy the requirements of §§ 25-7-102, -105(1)(e), -106, and/or -109, C.R.S., as applicable.

- (III) Evidence in the record supports the finding that the rule shall bring about reductions in risks to human health and the environment that will justify the costs to government, the regulated community, and to the public to implement and comply with the rule.
- (IV) The rules are the most cost-effective to achieve the necessary and desired results and reduction in air pollution.
- (V) The rule will maximize the air quality benefits of regulation in the most cost-effective manner.

Editor's Notes

History

New rule eff. 06/14/2023.

Entire rule, Part E IV eff. 12/15/2023.

Part A II.AA-II.QQQ, Part B II.A.5-II.A.6, III.B, IV.B.1.d, IV.C.1.d, IV.D.1.d, Part D, Part E V eff.
02/14/2025.